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School of Electrical and Computer Engineering

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Final Report

MagViVit: Transformer-based Spatiotemporal Magnification for Vital Signs Monitoring

Project number:	p-2026-106
Students (name & ID):	Almog Talker 322546680 Ilay Baruch 209128677
Supervisors:	Prof. Ofer Hadar Mr. Maor Oz
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Project repository: https://github.com/ilaybaruch10/Final_Project_MagVivit

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1 Project Closing Abstract

1.1 תקציר סיום פרויקט

ניטור רציף של מדדים חיוניים ביחידה לטיפול נמרץ יילודים (NICU) הוא הכרחי, אך הפרקטיקה הנהוגה מסתמכת על חיישנים דביקים הגורמים לנזקי עור (MARSI), לדחק נוירו-התפתחותי ולירידה באמינות עקב עיוותי תנועה (ארטיפקטים). מטרת הפרויקט הייתה לפתח מודל למידה עמוקה כהוכחת היתכנות (Proof of Concept) לניטור קצב נשימה (RR) ללא מגע מתוך וידאו NIR בלבד.

במהלך הפרויקט הוקמה והופעלה מערכת איסוף נתונים קלינית מלאה במחלקת ה-NICU בבית החולים "סורוקה" בבאר שבע, לאחר קבלת אישור ועדת הלסינקי (IRB). המערכת כוללת מצלמת NIR אחת של e-con Systems המכוונת אל הפג ומצלמות RGB המכוונות אל צג המוניטור, יחידת קצה NVIDIA Jetson AGX Orin, כונן NVMe בנפח 1TB, זרועות ומתאמים בהדפסת תלת-ממד, ותהליך הקלטה (Pipeline) מבוסס GStreamer המבטיח הקלטה מסונכרנת ללא אובדן פריימים בקצב 30 fps. במקביל, פותח תהליך עיבוד תמונה קלאסי לחילוץ אמת-מידה (Ground Truth) מצג המוניטור הרפואי: זיהוי מרובע המסך, מעקב נקודות בשיטת KLT, אמידת הומוגרפיה ב-RANSAC, החלקת פינות באמצעות מסנן קלמן, יישור פרספקטיבי וקריאת הספרות (OCR). המערכת אספה 500 קטעי וידאו מתשעה פגים.

הארכיטקטורה הראשונה שנבנתה, בהתאם למפרט המקורי, הייתה רגרסיה צפופה מבוססת מודל ResNet-50 קפוא ומקודד Transformer. היא נכשלה, והכשל נמדד: מאפייני ה-ResNet לאחר מיצוע מזהים את זהות הפג בדיוק של 96.3%, ופיזור החיזויים עבור כל פג בנפרד קרס ל-0.3 עד 2.1 נשימות לדקה. המודל למד לזהות מיהו התינוק, ולא כיצד הוא נושם. ניתוח שורש הבעיה חשף את המנגנון: כל מיצוע, מרחבי או על פני 2048 ערוצי המאפיינים, מדלל את אות הנשימה – שתופס חלק קטן מהתמונה ומופיע במעט ערוצים – אל מתחת לרמת בלוקי הפיקסלים הגולמיים ב-25 מתוך 27 הקטעים שנדגמו. מדידה של כל ערוץ בנפרד מראה שהקצב דווקא קיים במאפיינים, ולכן הכשל נבע ממיצוע המידע ולא מהיעדרו.

בעקבות זאת שונה כיוון הפתרון לייצוג בתחום התדר. כל חלון של 10 שניות (בפסיעה של שנייה, 11 חלונות לקטע) עובר הגברת תנועה נלמדת (מקדם 8, מסנן Butterworth בתחום 0.2 עד 2.0 הרץ), התמרת פורייה על בלוקי פיקסלים בשלוש סקאלות, הלבנה ספקטרלית, ודגימה מחדש על רשת תדרים לוגריתמית (160 תאים, 10 עד 200 נשימות לדקה). רשת קונבולוציה חד-ממדית בעלת ארבע שכבות ושקילות-הזזה מפיקה התפלגות על התאים, ותוחלתה היא קצב הנשימה. הבחירה בציר לוגריתמי היא מהותית: הכפלת הקצב בקבוע הופכת להזזה טהורה, והקריאה היא בעלת שקילות-הזזה מעצם בנייתה. כך, הממדים המרחביים שדרכם דלפה זהות הפג בארכיטקטורה הראשונה נעלמים מן הקלט. שלוש מדידות מגדירות כעת מה הושג בכך ומה לא. פיזור החיזויים עבור כל פג בנתוני המבחן עומד על 1.5 עד 9.3 נשימות לדקה במערכת החדשה, לעומת 0.3 עד 2.1 בארכיטקטורה הקודמת, כך שהקריסה נפתרה. עם זאת, מסווג לינארי עדיין מזהה את הפג מתוך הספקטרום ב-85.5% מקטעי המבחן, לעומת רצפת ניחוש של 18.5%, ולכן מידע הזהות שרד בקלט: המשמעות היא שהקריאה מונעת את העברתו אל הפלט, ולא שהוא הוסר לחלוטין. השמטת שלושת הערוצים הנגזרים מ-ResNet אינה משנה את הדיוק במידה הניתנת למדידה.

תוצאות אימות בשיטת leave-one-subject-out על תשעה פגים (27 מודלים, תשעה קיפולים בשלושה זרעים אקראיים): שגיאה מוחלטת ממוצעת ברמת הקטע של 9.73 נשימות לדקה, עם רווח סמך 95% של [6.9, 12.9], לעומת 13.6 עד 14.0 עבור חיזוי חסר-מידע. המערכת מציגה ביצועים טובים יותר מגלאי שיא קלאסי ב-2.58 נשימות לדקה, עם רווח סמך [0.4, 5.2], בשמונה מתוך תשעה פגים. מקדם המתאם התוך-נבדקי הוא 0.288, עם רווח סמך [0.123, 0.409], שאינו כולל אפס. אימון למספר תקופות קבוע של 150, ללא כלל עצירה מוקדמת, מניב 9.26 נשימות לדקה, עם רווח סמך [6.6, 12.4]. ההפרש אינו מובהק סטטיסטית, ולכן התוצאה אינה תלויה בכלל העצירה. רצפת רעש התיוג עצמה היא 3.2 נשימות לדקה.

מסקנה: הוכחה היתכנות הנדסית של שרשרת העיבוד המלאה, והודגם כי ניתן לשחזר את האות הנשימתי מתוך וידאו NIR בלבד. כל רווחי הסמך חושבו באמצעות דגימה חוזרת של פגים ולא של קטעים. הדיוק הנדרש לשימוש קליני נותר יעד להמשך הדרך, ובמדגם של תשעה פגים, ההפרש הקטן ביותר שניתן להבחין בו הוא 2.17 נשימות לדקה.

מילות מפתח: טיפול נמרץ יילודים (NICU), ניטור ללא מגע, למידה עמוקה, הגברת תנועה בווידאו, וידאו NIR, ניתוח ספקטרלי, ציר תדר לוגריתמי, שקילות-הזזה, קצב נשימה, אימות leave-one-subject-out.

1.2 Abstract

Continuous vital-sign monitoring in the Neonatal Intensive Care Unit (NICU) is essential, yet standard practice relies on adhesive sensors that cause skin injuries (MARSI), neurodevelopmental stress and frequent data loss through motion artifacts [1], [2]. Our goal was a proof-of-concept deep-learning system that estimates respiratory rate (RR) contactlessly, from near-infrared (NIR) video alone.

We designed, built and deployed a complete clinical acquisition system in the NICU of Soroka Medical Center, Beer-Sheva, under Helsinki Committee (IRB) approval: one e-con Systems NIR camera on the infant and further RGB cameras on the bedside monitor, an NVIDIA Jetson AGX Orin edge unit, a 1 TB NVMe SSD, 3D-printed articulating mounts, and a GStreamer pipeline recording synchronized, frame-loss-free video at 30 fps. A classical computer-vision pipeline recovers ground truth photographically from the bedside monitor, through screen-quad detection, KLT tracking [3], RANSAC homography [4], Kalman-filtered corner smoothing [5], rectification and digit reading. The system has collected 500 clips from nine preterm infants.

The first architecture followed the original specification: dense per-frame regression through a frozen ResNet-50 and a Transformer encoder. It failed, and the failure is measured. Pooled ResNet features identify which infant is in the clip with 96.3 % accuracy and held-out prediction spread collapsed to 0.3 to 2.1 bpm within each infant: the model had learned to recognize the baby instead of the breathing. Any statistic averaged over the region, or across the 2048 feature channels, dilutes the respiratory rate below the level of raw pixel blocks on 25 of 27 sampled clips. Measured per channel the rate is present, so the defect is the pooling of the evidence and not its absence.

We therefore moved the read-out into the frequency domain. Each 10 s window (1 s hop, 11 per clip) is magnified (learning-based video motion magnification [6], amplification 8, Butterworth 0.2 to 2.0 Hz), reduced to pixel-block power spectra at three block sizes, whitened per block, and resampled onto a logarithmic rate axis of 160 bins covering 10 to 200 bpm. A four-layer shift-equivariant 1-D CNN produces a distribution over those bins and its expectation is the estimated rate. The logarithmic axis is the essential choice: multiplying the rate by a constant becomes a pure translation, so the read-out is shift-equivariant by construction and the spatial axes the first architecture exploited are gone from the input. Three measurements define what that achieves. Within-infant prediction spread is 1.5 to 9.3 bpm against 0.3 to 2.1 for the design it replaced, so the collapse is gone. A linear classifier still names the infant from the spectra on 85.5 % of held-out clips against an 18.5 % floor, so identity survives in the input and the defensible claim is that the read-out cannot route it to the output. And dropping the three ResNet-derived channels changes clip error by -0.01 bpm, CI [-1.5, +2.0].

Under nested leave-one-subject-out validation over nine infants, nine folds by three seeds and 27 models, with every infant held out exactly once and no clip of a test infant in the training or validation set of its own fold, clip-level mean absolute error is 9.73 bpm, 95 % CI [6.9, 12.9], against 13.6 to 14.0 bpm for a no-skill constant predictor. The system beats a calibrated classical peak detector by 2.58 bpm, CI [0.4, 5.2], on eight of nine infants. Within-subject correlation is 0.288, CI [0.123, 0.409], excluding zero, so the model tracks rate within an infant and not only between infants. A fixed 150-epoch budget, which ignores the validation infant entirely, gives 9.26 bpm, CI [6.6, 12.4], and an independent repeat gives 9.25, so the headline does not depend on the stopping rule. Clip RMSE is 12.5 bpm, an MSE of 157 bpm^2, against a declared target of 2.0, so the shortfall on the metric as declared is larger than the mean absolute error suggests. The reference itself has a within-window noise floor of 3.2 bpm and trails the monitor display by a measured half second.

Engineering feasibility of the whole chain was demonstrated. Every interval above is a bootstrap resampled over infants and not over clips, because clips from one baby are not independent observations. The delivered read-out has 29,537 trainable parameters against 19,031,809 in the design it replaced and the whole nine-fold, three-seed protocol retrains in 5.7 minutes on the Jetson, so every number here is

cheap to regenerate. The accuracy required for clinical use remains a target for continuation, and at nine infants the smallest difference this cohort can resolve is 2.17 bpm.

Keywords: Neonatal Intensive Care Unit (NICU), contactless monitoring, deep learning, video motion magnification, NIR video, spectral analysis, logarithmic frequency axis, shift equivariance, respiratory rate, leave-one-subject-out validation.

2 Introduction

2.1 Clinical motivation

Respiratory rate is one of the most informative vital signs in the neonatal intensive care unit, and apnoea of prematurity makes continuous monitoring of it mandatory and not optional [7], [8]. The standard means is an adhesive chest lead, which causes medical adhesive-related skin injury in a population whose epidermis is not yet a barrier [1], [2], contributes to neurodevelopmental stress [9], and loses data to motion artifacts [10], [11]. A camera introduces no contact at all, which is why non-contact neonatal monitoring has been an active field for a decade.

A camera introduces no contact at all, which is why non-contact neonatal monitoring has been an active research area for over a decade [12], [7], [8], [13], [9], [1], [10], [11]. It is also much harder than the adult case:

- **Signal amplitude.** The chest displacement of a preterm infant during quiet breathing is frequently sub-millimeter, close to or below the noise floor of a conventional camera.
- **Motion artifacts.** Infants move spontaneously and erratically, producing displacements one to two orders of magnitude larger than the respiratory signal.
- **Environment.** Incubator walls, blankets, phototherapy lamps, ventilator tubing and caregiver interventions occlude the region of interest for long stretches, and the ambient illumination changes between day and night shifts.

2.2 Project goal

The goal was a proof-of-concept software model that ingests NIR video of an infant in an incubator and outputs a continuous estimate of respiratory rate and body temperature, validated against ground truth read from the bedside clinical monitor. The targets are MSE at most 2.0 breaths/min and MSE at most 0.2 °C for temperature. The goal has two halves, building the clinical acquisition and ground-truth system and building a model that turns its output into vital signs; the first was delivered in full.

2.3 Alternative approaches and why ours was chosen

Three families of method are used for non-contact neonatal respiratory-rate estimation, and each bore on this project differently.

- **Classical signal processing.** Eulerian magnification [14], phase-based processing [15] and spectral peak finding amplify or track intensity variation in a region of interest, then take the dominant frequency by FFT or band-pass filtering. No training data is required, the computation is cheap and the result is interpretable. This is the family the delivered system belongs to, and its weakness, that it has no mechanism for rejecting artifact intervals, is handled here by learning the read-out while the representation stays fixed.
- **Probabilistic and state-space methods.** Bayesian trackers and Kalman filters [5] fit a dynamic model to the respiratory waveform and infer the rate from the posterior, which gives a principled treatment of uncertainty and degrades gracefully over short gaps. We use this family inside the ground-truth pipeline and not for the rate itself: a Kalman filter is run over the tracked screen-corner coordinates of the bedside monitor, with outlier rejection, in order to suppress the per-frame jitter in those coordinates while still following a genuine camera nudge, so that the rectified digit crop stays registered (§4.3).
- **Deep learning.** CNN, 3D-CNN, CNN plus LSTM and transformer models [12], [13], [16], [17], [18] learn the mapping from pixels to rate directly from labeled video. They are robust to illumination change and can learn to ignore non-respiratory motion, and they hold the best published results on

real infant video: a mean absolute error of about 2.9 bpm on the AIR-125 benchmark [12], and about 1.6 bpm on thermal video [13]. We tried this family first, in the form the specification described, and it failed by memorizing infant identity (§5.3). The lesson was that with nine infants the capacity has to be spent on the read-out and not on the representation.

The delivered design is a hybrid of the first and the third family, and the way the two are combined is the project's central engineering claim:

1. **Stage 3** uses a pretrained magnification network as a fixed, physically motivated preprocessor, so that sub-pixel chest motion is lifted above the sensor noise floor before any learning happens. This buys the sensitivity of the classical Eulerian methods without asking a regressor to discover amplification from 500 clips.
2. **Stage 4** throws away everything except the rate. Reducing each window to a power spectrum on a logarithmic rate axis discards spatial appearance entirely, and that is what removes the shortcut the first architecture had exploited.
3. **Stage 5** learns only where the peak is. A shift-equivariant 1-D CNN over the frequency axis has, by construction, almost no representable function other than "find the peak and report its position". General features transfer and task-specific layers do not [19], and at this data scale the honest move is to give the model as little freedom as possible.

2.4 Where this sits against published work

Table 1: *Published non-contact respiratory-rate accuracy, and this work. All figures are mean absolute error in breaths per minute.*

System	Modality	Cohort	MAE [bpm]
AIRFlowNet on AIR-125 [12]	RGB video, infants	125 videos, public benchmark	about 2.9
Adult-designed models on AIR-125 [12]	RGB video, infants	the same benchmark	4.7 to 6.2
Spatio-temporal deep learning [13]	Thermal video, neonates	not public	about 1.6
This work	NIR video, preterm infants	9 infants, 500 clips, LOSO	9.73
No-skill predictor on this cohort	none	9 infants	13.6 to 14.0

Our result is well short of the published state of the art, and the gap to close is the 2.9 bpm reported on AIR-125. What differs is the recording condition and the reference. AIR-125 is RGB video annotated for respiration from the video itself; ours is near-infrared video of preterm infants inside incubators, scored against a photograph of a bedside monitor whose own reading moves by 3.2 bpm inside a 10 s window (§4.3.2). A published benchmark and a ward reference are not the same measurement problem, and §7.7 recommends evaluating on AIR-125 precisely so the two can be placed on one axis. The length of the interval each estimate is computed over differs too. Ours is a single 10 s window inside a 20 s clip, which holds five to ten breathing cycles at 30 to 60 bpm, and an estimate formed over a longer recording averages more cycles and is correspondingly less sensitive to any one irregular breath. To the extent that the published figure is computed over longer videos than ours, part of the gap follows from the analysis interval alone.

3 Final Technical Specification

3.1 System-level specification

Table 2: *Final system-level specification of the delivered proof-of-concept.*

Item	Final value (as built)
Deployed input modality	NIR video, magnified
Analysis unit	10 s window, 1 s hop, 11 windows per 20 s clip
Acquisition rate	30 fps, synchronized, no frame loss (GStreamer)
Magnification	Learning-based video motion magnification [6], amplification 8, Butterworth band-pass 0.2 to 2.0 Hz
Representation into the model	`(6, 160)` power spectrum on a logarithmic rate axis
Rate range covered	10 to 200 breaths/min across 160 bins
Model	4-layer shift-equivariant 1-D CNN over the frequency axis
Primary output	One RR estimate per 10 s window
Secondary output	Clip-level RR, the mean of the 11 window estimates
Temperature output	Not implemented
Operating mode	Offline, on recorded clips
Ground-truth reference	Digits read from the bedside monitor by a CV pipeline (§4.3)
Validation protocol	Nested leave-one-subject-out, 9 folds by 3 seeds
Reported metrics	MAE with infant-clustered bootstrap CI, Pearson r overall and within-subject, Bland-Altman bias and limits of agreement, no-skill baseline
Measured clip MAE	9.73 breaths/min, 95 % CI [6.9, 12.9] (nested early stopping). 9.26 with a fixed 150 epochs, CI [6.6, 12.4]
Acquisition hardware	e-con Systems cameras on the Sony IMX485 sensor (e-CAM82 CUOAGX, 4-lane MIPI CSI-2, `e-con_cam` driver 1.1.0-ge3f455), NVIDIA Jetson AGX Orin developer kit (p3701-0000 on p3737-0000), 1 TB NVMe, JetPack 5.1.1 (L4T R35.3.1)

3.2 Signal representation

The step that distinguishes this system from the one originally specified is what the network is shown. Nothing spatial survives.

Table 3: *Construction of the* `(6, 160)` *input, per 10 s window.*

Element	Value	Purpose
Window length	10 s at 30 fps	Long enough for several breathing cycles at 30 to 60 bpm, so a spectral peak is defined at all (§4.7)
Hop	1 s, giving 11 windows per clip	Averaging the 11 estimates absorbs per-window variance
Region of interest	192 x 192 px torso crop	Set in `blockdet.py`. Everything downstream is computed on this crop, not on the full frame
Order statistics	max, p90 and mean over the pooled 336 blocks	Robust pooling. A plain mean over the frame is what diluted the signal in the first architecture (§5.3)
Block scales	12, 24 and 48 px, giving 256 + 64 + 16 = 336 block time series, pooled	Three grids so the chest is covered whatever its apparent size and position
Channels	six, one per order statistic	Three order statistics over the pooled 336 pixel blocks, then the same three over the 2048 cached ResNet-50 [20] feature channels, frozen and pretrained on ImageNet [21]. A channel is a statistic, not a scale. No ResNet runs inside the training loop
Detrending before the transform	Per-block mean removed, then a linear and a quadratic basis projected out; Hann window; transform zero-padded to 16 times the window length	A drift or a slow curve across the window would otherwise put its energy in the lowest bins. The zero padding interpolates the grid so the log resampling has something to resample
Spectral whitening	Least-squares fit of log P = a + b log f over 5 to 280 bpm, divided out, each spectrum then normalized to sum 1	Removes the 1/f background so the respiratory peak is judged against a flat floor. Normalizing to unit sum discards amplitude, so how strongly the chest moves cannot reach the read-out, only at what rate
Frequency grid	160 bins, logarithmic, 10 to 200 bpm, deliberately wider than the	A change of rate becomes a translation (§4.7). Each bin is a factor of 1.019, and the reference range of 19.5 to 97.5 bpm

	label range	lands between bins 36 and 122, so the peak never sits against an edge of the grid
Final scaling	Each channel divided by its own maximum, then log(x + 1e-6)	Puts the six channels on a common scale and turns a multiplicative peak-to-floor ratio into an additive one

3.3 Model specification

Table 4: *The delivered model,* `ShiftEquivariant` *in* `tools/train_logfreq.py`.

Component	Configuration
Input	`(6, 160)`
Trunk	4 convolutional layers, `Conv1d`, width 32, kernel 9, same padding throughout
Equivariance	Same padding and no pooling or stride, so a translation of the input produces the same translation of the output
Head	One logit per frequency bin, then softmax over the 160 bins
Read-out	Expectation of the resulting distribution over the logarithmic rate axis
Loss	Soft-target cross-entropy over frequency bins, the target being a Gaussian centered on the reference rate. Not a Huber loss on bpm [22]
Augmentation	Translation along the frequency axis, which is a change of rate and therefore label-consistent by construction

The whole trunk is small on purpose. Four same-padded convolutional layers of width 32 and kernel 9 over a six-channel, 160-point input come to 29,537 trainable weights, counted from the instantiated model.

3.4 Cohort and validation protocol

Table 5: *Cohort and validation protocol.*

Property	Value
Infants	9
Clips in the clean cohort	500 (`data/metadata_9inf_dedup.csv`)
Clips entering training	496. All 500 have cached features; four drop out because every one of their windows fails the half-labelled rule, their valid-frame fractions running from 0.065 to 0.223
Windows	5,431. A 20 s clip yields 11 windows at 10 s length and 1 s hop, so 496 clips would nominally give 5,456. In practice 492 clips give 11 windows, one gives 10 and three give 3, because a window is discarded unless at least half of its frames survive the label mask
Subject definition	One recording date is one infant. `Patient_0` on two dates is two babies. One infant has two records only where a patient folder holds both `Record_0` and `Record_1`
Split	Nested leave-one-subject-out. Per fold and seed: 7 train, 1 validation used only to choose the stopping epoch, 1 test
Models trained	9 folds by 3 seeds = 27. Seed distributions are averaged before read-out
Stopping rule	Early stopping on the validation infant, median selected epoch 11

3.5 Compute and runtime

Table 6: *Compute environment.*

Item	Value
Device	NVIDIA Jetson AGX Orin [23]
Environment	`/mnt/pag_disk/rr_venv`, with `LD_LIBRARY_PATH=/mnt/pag_disk/rr_venv/syslibs`
Feature caching	ResNet features and spectra are computed once, offline, and cached with PyTorch [24] and the TorchVision IMAGENET1K_V2 weights [25]. Training reads spectra, not video
Cached-spectrum build	3.48 s per clip at the median (1.11 s to read 600 PNG frames, 0.11 s for the 336 block series, 2.20 s for the whitened spectra and order statistics of all 11 windows), so about 29 min for the 500-clip cohort
Training cost	0.31 s per epoch on the Orin GPU over 3,583 training and 902 validation windows. A fold-seed runs to `best_epoch + patience + 1`, a median of 37 epochs, so 11.4 s at the median and 8.0 to 18.8 s across the 27. All 27 fold-seed models train in 340 s

Whole LOSO protocol	5.7 min with nested early stopping; 20.8 min for the fixed-150-epoch configuration (27 x 150 x 0.31 s)
Priority policy	The ward acquisition service outranks training. The training job is deprioritized with `renice` and `ionice`, never the other way round

3.6 Changes with respect to the preliminary specification

Each row gives the change and the reason, and the reasons are developed in §4 and §5.

Table 7: *Differences between the preliminary specification and the delivered system.*

Item	As originally specified (preliminary report and supervisor specification)	Delivered	Reason
Deployed input	Fused RGB + NIR	NIR only	Removes runtime inter-camera registration. RGB is needed only to photograph the monitor.
Representation	Video frames, 224x224 in the supervisor specification; we ran the pipeline at a larger crop	Power spectrum on a logarithmic rate axis, `(6,160)`	Spatial appearance is the channel through which identity leaked (§5.3). Removing the spatial axes removes the route the first architecture took, though §5.5.2 measures that it does not remove the information.
Temporal model	6-layer, 8-head Transformer	4-layer shift-equivariant 1-D CNN	A translation-equivariant read-out over frequency can express little except peak-finding, which is what we want it to do.
Backbone	Frozen ResNet-50 inside the loop	ResNet features cached offline, spectra only	The features carry less energy at the true rate than raw pixels on 25 of 27 sampled clips (§5.3), so they are a secondary channel, not the primary one.
Output granularity	Dense, 200 values per clip	One value per 10 s window, 11 per clip	The monitor label has no per-frame resolution to supervise against, median SD 3.2 bpm within a 10 s window, so dense supervision was partly fictional (§4.7).
Loss	Masked Huber on bpm + smoothness	Soft-target cross-entropy over frequency bins	Predicting a distribution over rate matches both the physics and the label uncertainty better than a point regression.
Validation	A fixed 43/10/10 clip-level split inherited from the legacy split column of the early metadata file, which covered one infant's recordings, so the subject grouping it specified had no effect	Nested leave-one-subject-out, 9 infants, 27 models	With nine infants a single held-out fold is too small to mean anything. LOSO uses every infant as a test subject once.
Uncertainty reporting	Not specified	Bootstrap CIs clustered on infants, no-skill baseline beside every MAE	Clips within one baby are not independent, and an MAE without a baseline cannot be interpreted.
Augmentation	Extensive: spatial, temporal, synthetic noise and motion artifacts	Translation along the frequency axis only	The spectral representation has no spatial or temporal axes left to augment. Translation along the rate axis is the only label-consistent transformation available to us, and unlike the planned ones it is exact and not approximate (§4.7.2).
Temperature	MSE <= 0.2 °C, trained jointly	Not implemented	The monitor displayed temperature for too few of the recorded infants, and where it was displayed it was sometimes on a separate monitor that would have needed another camera, so there was no reference to train against.

4 Approach to the Solution and Engineering Design

4.1 System overview

The system is a five-stage pipeline. Stages 1 and 2 build the dataset, stage 3 amplifies the motion, stage 4 converts a window of video into a rate spectrum, and stage 5 reads the rate off that spectrum. The stages communicate through files on disk, so each one can be re-run and audited independently.

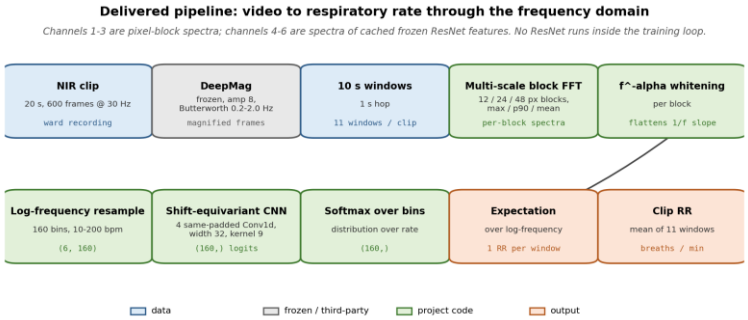


Figure 1: *The delivered pipeline. Gray is frozen or third-party, green is code written for this project, blue is a data artifact, orange is a result. Every spatial dimension is gone by the end of the first row, by design.*

Because the stages communicate through files, every intermediate product of pre-processing is a named artifact that can be inspected and re-run on its own. The chain below is the one every number in chapter 5 was computed through; §3.2 specifies the content of each artifact, §4.5 the arithmetic that produces it, and Appendix A gives the full repository map.

```
Test_Records_ROIs/<date>/Patient_N/Record_M/vid_K/    stage 1: ward capture
    motion_frames/       torso crop, one PNG per frame, 30 fps
    chromatic_frames/    RGB crop, unused by the delivered path
    signs/rr.npy         per-frame reference rate read off the monitor (stage 2)
         |
         |  build_metadata_from_magnified.py   label QC: NaN, stale, out-of-range
         v
data/metadata_9inf_dedup.csv                   the cohort: 500 clips, 9 infants
         |
         |  deep_motion_mag (third-party, TF 1.x)   amp 8, Butterworth 0.2-2.0 Hz
         v
deepmag_work/<video>/                          magnified frames (stage 3)
         |
         |  tools/cache_windowed.py            blocks, FFT, whitening, log resample
         v
data/winspec_9inf.npz + winspec_9inf_index.csv 5,431 windows of (6,160)  (stage 4)
         |
         |  tools/train_windowed.py            LOSO, 9 folds x 3 seeds  (stage 5)
         v
runs/<name>/pred_noaug.csv, pred_windows.csv   stored predictions, scored by
                                               tools/review_stats.py
```

One property of that chain matters for reading chapter 5: the expensive stages, magnification and spectrum caching, run once and write files, after which the model trains from `winspec_9inf.npz` in minutes. That is why the read-out experiments of §5.5 and §5.5.2 could be run at all, and why §7.7 calls a sweep over magnification settings cheap and a new architecture expensive.

4.2 Stage 1: clinical acquisition

Recording infants in an active intensive-care unit imposes constraints that dominated the first half of our work. The unit cannot be modified, staff workflow must not be obstructed, and nothing may touch the patient.

- **Ethical approval.** Obtained from the Helsinki Committee (IRB) before any recording, as required for research on human subjects in Israel.
- **Recording sessions.** Nine dated sessions were recorded and are on disk: 2025-12-31, 2026-01-21, 2026-04-15, 2026-04-29, 2026-05-14, 2026-05-27, 2026-06-04, 2026-06-17 and 2026-06-24,

holding 601 video directories between them. The analyzed cohort draws on eight of those dates and nine infants, because 2026-05-27 contributed two patients. Two sources are on disk and unused: the whole of 2026-04-29, 63 clips from one patient, and the second patient folder of 2026-06-17. Both were left out at the deduplication stage and neither was excluded on the basis of any result. §7.7 lists their recovery among the continuation tasks.

- **Sensing and compute.** One e-con Systems camera observes the infant inside the incubator, and the further RGB cameras aimed at the vital-signs monitor supply the ground truth. All connect to a single NVIDIA Jetson AGX Orin developer kit with a 1 TB NVMe SSD. The camera identification is taken from the flashed system image, the authoritative record of what the rig was built with: the `e-con_cam` driver, version 1.1.0-ge3f455, and the compiled device tree identify the sensor as a Sony IMX485 in e-con's e-CAM82_CUOAGX product tree, at four MIPI CSI-2 lanes per port, on p3701-0000 over p3737-0000. Two limits: it is a device-tree identification and not a label read off the hardware, so an exact ordering code needs the purchase order; and the device tree provisions four such modules, which is what the multi-camera image declares and is not proof that four were connected in any session.
- **Mechanical integration.** We designed and 3D-printed articulating arms and connectors so that the cameras clamp above the incubator and the monitors, hold their pose between sessions, and leave the clinical working volume free. Mechanical stability is not cosmetic: a vibrating mount produces exactly the low-amplitude periodic displacement that stage 3 is built to amplify, and stage 4 would read it as breathing.
- **Capture.** A GStreamer-based pipeline records all streams at 30 fps with no frame loss and a common time base. This replaced an initial Python and OpenCV implementation that dropped roughly one frame in five (§6, problem 1).
- **Clinical documentation.** Each session is entered in a structured protocol mapping video files to gestational age, weight, skin tone, red-blood-cell count, treatment and blanket state. Skin tone and gestational age are recorded so that bias across the cohort can be measured later instead of assumed away.
- **Region of interest.** The ward recording service writes a torso crop for each clip, and that crop is what the spectral front end consumes. We also annotated two classes, face and face plus torso, in CVAT [26] in YOLO format, intending to train a detector with SORT tracking [27] as the preliminary report described. The annotation was completed; the detector was never trained, and §7.7 lists it as the first pre-processing task for a successor.

4.3 Stage 2: Ground Truth from the bedside monitor

Ground Truth is the reading already displayed on the certified clinical monitor, recovered photographically. This avoids attaching any additional instrument to the infant, and our reference is the same number the clinical staff act on. The cost is that the reference has an error chain of its own, which has to be characterized.

The monitor is photographed obliquely by a fixed RGB camera, so recovering legible digits requires geometric normalization: intrinsic calibration and lens undistortion; contour-based detection of the screen quadrilateral; KLT optical-flow tracking of the corners [3] so a momentary occlusion does not lose the screen; RANSAC homography estimation [4], which rejects corners displaced by reflections; Kalman filtering of the corner trajectories [5] to suppress per-frame jitter while still following a genuine camera nudge; then perspective rectification and digit reading. The reader is stabilized by a temporal voter that returns the mode of its last 30 successful reads.

4.3.1 Three independent label failure modes

Instead of discarding a clip that contains some bad labels, `build_metadata_from_magnified.py` emits a per-frame validity mask alongside the label array. Three failure modes are detected, and no single check finds the other two.

Table 8: *Ground-truth label failure modes detected by the label QC stage.*

Failure mode	Why it happens	How it is detected
Leading NaN	The voter returns nothing while its buffer is still empty, and the upstream interpolation never back-fills a leading gap.	A NaN test on the raw label array.
Stale voter output	Across a long run of failed reads the voter keeps returning the mode of its last 30 successful reads. The value is not NaN, it is fabricated, so no NaN test can find it.	Frames more than 30 behind the last successful read, counted from the debug crops.
Dropped digit	The OCR reads 40 as 4, or 50 as 5. The read succeeded, so the first two checks pass it through.	A physiological range test, 15 to 120 bpm. The lower bound sits below any rate a neonatal monitor would genuinely display and above the dropped-digit values, which are the leading digit of a real rate and so land in the single digits or low teens.

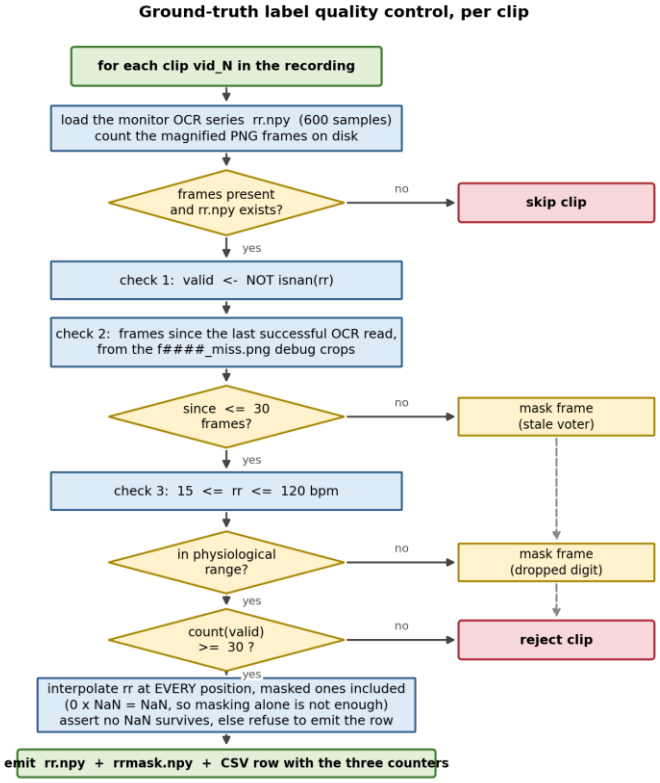


Figure 2: *Flow chart of the label quality-control logic, one pass per clip. The three diamonds are checks 2 and 3 of the table above, while check 1, the NaN test, is the process box above them. A frame that fails any of the three is masked without the whole clip being discarded, and a clip is rejected only when fewer than 30 frames survive.*

Labels are NaN-filled even at masked positions, because IEEE arithmetic gives 0 times NaN equals NaN, so masking alone would not keep a NaN out of the optimizer. A hard guard re-checks the filled array and refuses to emit the row if any NaN survives. The original label file is never modified, since it is shared with the upstream extraction pipeline.

A spike filter would not have caught the dropped-digit mode. Counting 47 as a plausible rate is correct in isolation; it is wrong only because the monitor was showing 47 for the ninth consecutive second while the display had moved on. Detecting it needs the stall length, not the value, which is why the three tests are independent and why none of them is a smoothness prior. Nothing is interpolated across a masked

stretch, because an interpolated label is a fabricated measurement, and a window is dropped outright if fewer than half its frames survive.

4.3.2 The resolution and timing of the reference

Two properties of this reference govern the whole design of the read-out.

- **Resolution.** The monitor does not report a new number every frame. It displays a smoothed estimate on its own internal cadence, and within a single 10 s window the monitor's own RR has a median standard deviation of 3.2 bpm. Across a whole 20 s clip the label's own standard deviation has a median of 5.1 bpm over the 500 clips, and exceeds 5 bpm on half of them, which is a swing of roughly 18 bpm inside one clip. There is therefore no per-frame truth to supervise against, and §4.7 draws the consequence.
- **Timing.** The stored label trails the number a nurse would have seen. We measured it by comparing change-points in the monitor's own digit crop against change-points in the label array, and the lag is -15 plus or minus 1 frames, half a second, constant on eight of the nine infants. It is 5 % of a 10 s window and therefore small here, but it would have been a 15-frame systematic misalignment for the per-frame output originally specified, so the windowed design absorbed a defect the original would have inherited. One infant is an outlier we flag instead of averaging away: `0617_P0` reports 28 label changes against 11 display changes and a best lag of -66 frames, and it is the same infant carrying 1,243 of the cohort's 1,731 out-of-range frames (§5.2), so both anomalies point at that infant's extraction and not at the method. What is still not measured is the value-level error, how often a wrong digit passes all three checks of §4.3.1 and enters the label as a plausible number.

4.4 Stage 3: learned video motion magnification

The chest displacement we need to measure is often sub-millimeter at the sensor. Instead of asking a network to discover amplification for itself, we use the pretrained MIT learning-based video motion magnification network [6] as a fixed preprocessor, in temporal-filtering mode. It is the successor to Eulerian video magnification [14] and phase-based processing [15]: the amplification is performed by a convolutional autoencoder trained on synthetic motion, which rings far less than the hand-designed pyramids.

The configuration is a Butterworth band-pass with cut-offs at 0.2 Hz and 2.0 Hz, a 30 Hz sampling rate, one filter tap, and amplification factor 8:

```
sh run_temporal_on_test_videos.sh \
   o3f_hmhm2_bg_qnoise_mix4_nl_n_t_ds3 \
   <video_name> 8 0.2 2.0 30 1 butter
```

The choice of a band-pass over the two-frame static or dynamic modes matters, because those modes amplify whatever differs between two frames, gross body motion included. A pass-band of 0.2 to 2.0 Hz covers the neonatal range of 0.5 to 1.5 Hz with margin while attenuating both the slow drift of a settling limb and the fast transient of a startle. It front-loads the job stage 4 does again in the frequency domain, which is deliberate redundancy: the band-pass removes the energy, the spectrum makes what remains legible.

We made a small modification to the third-party code, adding a `butterLowpass` filter type which, unlike the band-pass variants, retains the DC component of the amplified shape signal. It is a TensorFlow 1.x codebase whose environment is incompatible with our PyTorch package, so the two live in separate virtual environments.

A preprocessor this central has to be verified, never assumed. Provenance first: the folder the metadata points at differs from the folder fed to the magnifier on every clip sampled, so the trainer is not silently reading raw video. Selectivity is the interesting check. The network is learned and not a linear filter, so

some gain outside the pass-band is expected; what would be bad is a clip whose out-of-band gain exceeds its in-band gain, because that clip has had its noise and its gross body motion amplified harder than its breathing and is worse than the raw video it came from.

Measured on 61 clips, comparing median per-pixel spectral power before and after: in-band gain has a median of 4.47 and out-of-band gain 2.73, so the pass-band is amplified preferentially, and the ratio exceeds one on 45 of the 61 clips. On the other 16 it does not. Two facts limit what that supports. These clips come from the pre-deduplication metadata, so they cover eight of nine infants with one infant under two labels. And the two least selective infants, `0121_P0` at 0.61 and `0624_P0` at 0.82, are not the two worst folds in §5.5, scoring 7.75 and 8.69 bpm against a cohort mean of 9.73. So the stage measurably does what it should on roughly three quarters of clips, and we cannot show the remaining quarter is what limits accuracy; that needs the ablation §7.7 recommends.

Two published results say this is expected behaviour and not a fault of our footage. The authors of the magnification network state that its quality degrades with temporal filters when the magnification factor is high and the motion small [6], which is exactly amplification 8 on sub-millimeter chest displacement. And Malik and Gulati ran the same pretrained network on neonatal bedside video and reported that for breathing the difference between original and magnified video is insignificant [28]. Our 16 clips of 61 are that limitation measured on our own data, which makes the stage a candidate for replacement and not for tuning (§7.7).

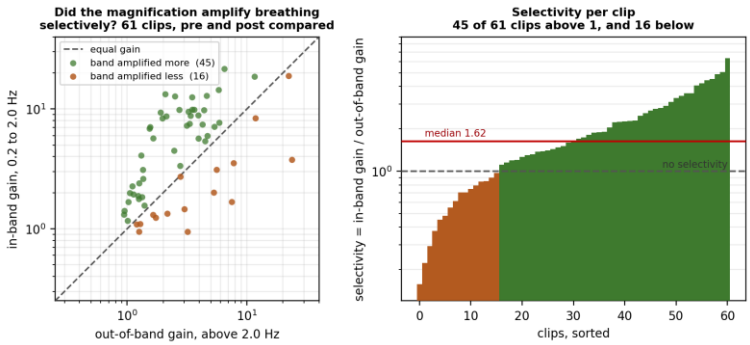


Figure 3: *Verification of the magnification stage. Left: for each of 61 clips, the gain inside the 0.2 to 2.0 Hz pass-band against the gain above it, both as ratios of median per-pixel spectral power after magnification to before. Points above the dashed line are clips where breathing was amplified more than everything else. Right: the same clips as a sorted selectivity ratio. The median is 1.62 and 16 clips fall below one, meaning the magnifier worked against us on those. This is a property of the pretrained network on our footage, not something we tuned.*

4.5 Stage 4: from video to a rate spectrum

This stage is the redesign. Its job is to destroy every property of the window except the rate.

1. The magnified window is divided into pixel blocks at three sizes, 12, 24 and 48 px. Each block yields one time series over the 10 s window.
2. Each block series has its mean removed and a linear and a quadratic basis projected out, is multiplied by a Hann window, and is Fourier-transformed to a power spectrum, zero-padded to 16 times the window length so the log grid has a finely sampled spectrum to resample from.
3. Each block spectrum is whitened by a fitted `f^-alpha` law. Natural video has a 1/f background, and without whitening the strongest bin is almost always the lowest one. After whitening a respiratory peak is judged against a flat floor.
4. The three scales are then pooled into a single set of 336 block spectra, 256 from the 12 px grid, 64 from the 24 px grid and 16 from the 48 px grid, and scale identity is not retained past that point. Across that pooled set we take three order statistics per frequency bin: the maximum, the 90th percentile and the mean. The maximum finds a peak present in only a few blocks, which is the normal case when only the chest moves; the mean is the robust fallback.

5. So each channel is one order statistic taken across the whole pooled set: channels 1 to 3 are max, 90th percentile and mean over the pooled 336 pixel-block spectra, and channels 4 to 6 are the same three statistics over the 2048 frozen ResNet feature channels, giving an input of `(6, 160)`.
6. Everything is resampled onto a logarithmic rate axis of 160 bins covering 10 to 200 bpm, each channel is divided by its own maximum, and the logarithm is taken.

Two rules govern which windows exist at all. A window is kept only if at least half of its 300 frames survive the label mask of §4.3.1, and its label is the mean of the surviving reference values over exactly those frames. Nothing is interpolated across a masked stretch and no window inherits a label from its neighbours.

This stage leaves a six-channel picture of how much periodic energy the window contains at each rate. What does not survive is where in the frame that energy was, what the infant looks like, or how bright the scene is.

4.6 Stage 5: the shift-equivariant read-out

The read-out is four same-padded `Conv1d` layers of width 32 and kernel 9, producing one logit per frequency bin, followed by a softmax over the 160 bins. The estimated rate is the expectation of that distribution along the logarithmic axis.

Two properties are enforced by the architecture, not learned. Because every layer uses same padding and there is no pooling or stride, translating the input spectrum translates the output distribution by the same amount. And because the input has no spatial axes, the spatial arrangement the first architecture exploited is not present: a read-out that cannot use absolute position along the frequency axis cannot emit a fixed per-infant constant, which is exactly what the first architecture did. There is a limit to the equivariance itself. Same padding makes it exact only away from the ends of the axis: with kernel 9 and four layers the receptive field is 33 bins, so the outermost 16 bins at each end, about 20 % of the axis, see the zero padding and could in principle expose absolute position. The clinically relevant range of 30 to 60 bpm sits in the interior.

What that argument establishes is narrower than removing identity, and §5.5.2 measures the difference. A physiological waveform is itself an identifier: the authors of [29] de-identify the facial video first, so identity can reach the classifier only through the recovered pulse waveform, and still authenticate subjects at a 0.17 % equal error rate from 20 heartbeats. A respiratory spectrum is coarser but not self-evidently anonymous, and it is not: a linear classifier names the infant from the (6, 160) spectra on 85.5 % of held-out clips against an 18.5 % floor. So the defensible claim is that the identity information survives in the input while the read-out is architecturally unable to route it to the output. Its legs are the equivariance argument above, under which a per-infant offset has no representation, and the measured symptom: within-infant prediction spread of 1.5 to 9.3 bpm against 0.3 to 2.1 for the design this replaced (§5.5.1). It also predicts that a less constrained read-out over these spectra would re-acquire the shortcut. The constraint is doing the work, not the representation.

The loss is a soft-target cross-entropy over the frequency bins, with the target a Gaussian centered on the reference rate, instead of a Huber loss on bpm. Three reasons. A distribution is the honest output for a quantity whose reference is itself uncertain by 3.2 bpm. A Gaussian target tolerates the harmonic structure of real breathing better than a point target. And cross-entropy over bins keeps the gradient informative when the prediction is far from the target, which is where the Huber loss of the earlier design had trouble.

Shift augmentation is free in this representation. Translating a spectrum along the axis is exactly a change of rate, so the augmented example carries a known, correct label. No other augmentation we could apply has that property.

4.7 Three design arguments

4.7.1 The 10 s window and the absence of a per-frame output

Breathing is cyclic, and a frequency cannot be measured from one sample. At 30 to 60 bpm a 10 s window holds five to ten cycles, which is the minimum for a spectral peak to exist at all; shorter windows blur the peak, longer ones average across genuine changes in rate. The 1 s hop then gives eleven overlapping estimates per clip, and averaging them absorbs per-window variance.

The specification asked for one estimate per frame, and we deliver eleven per clip. The reason is the label. The monitor updates its displayed rate every few seconds and its own reading moves by a median of 3.2 bpm inside a 10 s window (§4.3.2), so there is no per-frame ground truth to supervise against: a dense target built by holding the last displayed value would train the model to reproduce the monitor's update pattern, not the infant's breathing. We report this as the one requirement we do not meet, and §7.7 recommends revisiting it if a synchronized digital feed from the monitor is ever obtained.

4.7.2 The logarithmic rate axis

On a linear frequency axis, multiplying the rate by a constant moves the peak by a distance that depends on where it started, and widens it. A convolutional filter matched to a peak at 30 bpm is not matched to the same peak at 60 bpm. On a logarithmic axis it is a pure translation. Writing u = log f, a rate change f to kf maps u to u + log k, an additive shift whose size depends only on k and not on where the peak started; the peak keeps its shape and the second harmonic keeps a constant offset of log 2 from the fundamental. On our grid of 160 bins over 10 to 200 bpm each bin is a factor of 1.019, so a change of rate by half again is 21.5 bins wherever it occurs.

Convolution is exactly the operation that is equivariant to translation, so a same-padded 1-D CNN on this axis is matched to the family of all rates at once. Two consequences follow: the reason the redesign works:

- **Shift augmentation becomes free and exact.** A translated spectrum is a different rate with a known label, so the training set can be enlarged along the one axis that matters without any modeling assumption.

A shift-equivariant convolutional read-out over a logarithmic frequency axis with an expectation over a per-bin distribution is the architecture of PESTO for musical pitch, where the layers are sized so frequency resolution is never reduced [30]. Elowsson had already put a rate in beats per minute on a logarithmic axis for the same reason [31], and a distributional read-out over rate bins with a Gaussian soft target exists in this field already: PhysFormer casts heart rate as a 139-class problem on a linear axis [18]. What we contribute is the combination applied to respiration, and one argument for why respiration suits it better: our range of 10 to 200 bpm spans about 4.3 octaves where a heart-rate range of 42 to 180 spans 2.1, so logarithmic spacing buys more here, and the peak width in bins is constant across a range twice as wide.

- **The shortcut is closed by construction.** Find-the-peak-and-report-its-position is close to the only function a shift-equivariant network over this axis can represent. Spatial appearance, the channel the first architecture exploited, is absent from the input, and a read-out that cannot use absolute position cannot emit a fixed per-infant constant. §4.6 names the routes this does not close, and §7.7 gives the two tests that would settle them. The specific failure documented in §5.3, recovering the rate from spatial appearance, is not representable in this architecture at all.

4.8 Torso-motion measures and their spectra

A chest moving on screen is not the same thing as a respiratory rate a computer can measure, and the delivered front end is what came of testing that directly. The check is the following. Take a region of

interest covering only the part that clearly moves in the image, say only the torso or chest, and then compute mean absolute frame difference, vertical optical flow, ROI edge or contour displacement, the first temporal PCA component, and the mean motion-magnification residual. Then take the FFT or the PSD of those measures. The expectation is that the dominant motion frequency is approximately the ground-truth RR divided by 60. If the torso does not give a peak near that frequency, the problem is in the magnification, the preprocessing, or the labels.

We ran exactly that. The torso region occupies a median of 4.4 % of the frame. Each of the five measures reduces the window to one scalar time series, and each series was transformed and its dominant frequency compared against the reference.

Table 9: The torso-motion check executed on 27 clips spanning all nine infants.

Motion measure	Median rate error	Within 5 bpm	Median peak SNR
Mean absolute frame difference	20 bpm	7 of 27	7.5
Vertical optical flow	17 bpm	5 of 27	15.7
ROI edge displacement	18 bpm	6 of 27	6.8
First temporal PCA component	12 bpm	8 of 27	15.2
Mean magnification residual	12 bpm	8 of 27	16.3
Best of the five, per clip	7 bpm	11 of 27	
Disagreement among the five, per clip	48 bpm		

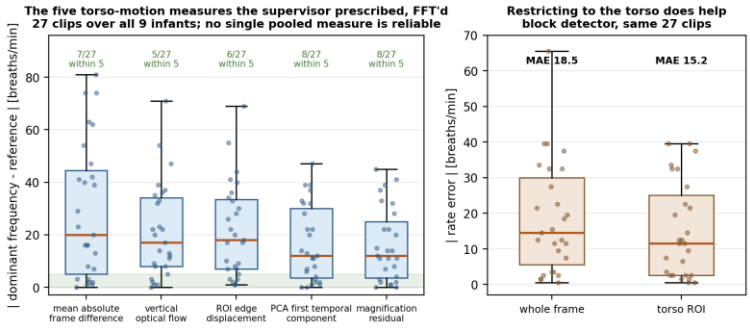


Figure 4: The torso-motion check, executed and measured. Left: for each of the five torso-motion measures, the distance *from its dominant FFT frequency to the reference rate, over 27 clips spanning all nine infants. The green band is agreement to within 5 bpm; no measure reaches it on more than eight of 27 clips, and on a given clip the five disagree with each other by a median of 48 bpm. Right: the same clips scored by the block detector on the whole frame and on the torso crop. Restricting to the torso does help, which is why the delivered front end keeps the crop.*

The premise of the check is right: respiratory energy is present in torso-motion measures, and the best of them lands its dominant frequency near the reference on a minority of clips. Median rate error across the five measures runs from 12.3 to 21.6 bpm, and the best, mean absolute frame difference, places 6 of 27 clips within 5 bpm. A pooled scalar per frame is not selective enough, which is the same finding §5.3 reaches from the other direction.

Wang and den Brinker set out camera-based respiratory motion extraction as algorithmic principles, and two of their conclusions are the ones this section reaches [32]. They segment the image into blocks, extract a respiratory signal from each, score each block by the ratio of its spectral peak to its total energy, and select blocks by that score. And they compare combining pixels before estimating motion against estimating motion per block and combining afterwards, finding the first clearly worse, because averaging first may fuse moving pixels with stationary ones or with pixels carrying a different motion source. That is the mechanism of §5.3, published in 2021. We reached it independently and by measurement, but it is a confirmation and not a discovery. The order statistics have precedent too: a mean over per-block spectra is the standard multi-channel periodogram, and a maximum over a per-block spectral map is how block selection is usually done.

Wang and den Brinker compare two pooling arrangements; we measure five spatial granularities, from a 16 by 16 grid to a single cell, on near-infrared neonatal footage, and report where the rate stops being recoverable at all, and not only which of two arrangements scores better. That curve, on this modality and this cohort, is the contribution.

The delivered front end is that check with the pooling deferred. Instead of one scalar per frame we keep 336, one per pixel block at three block sizes, and instead of choosing a measure in advance we transform all 336 and hand the read-out three order statistics across them. The maximum finds a peak living in a handful of blocks, which is the normal case when only the chest moves and is exactly what an average over the region dilutes away. The conclusion is therefore about where the pooling happens: a single pooled measure is not selective enough, and the fix is to pool last instead of first.

4.9 What the data preparation involved

§4.2 to §4.5 set out the four stages that turn a ward recording into a training example, and they set them out as a design. This section says what building them cost, because the chapter so far gives no sense of the proportions. The preliminary plan allowed fifteen weeks for data pre-processing and placed it in parallel with training; in practice the two ran in series, since there is nothing to train on until magnified clips and quality-controlled labels exist (§7.4). Almost every figure in chapter 5 was measured on a file produced by scripts we wrote, and the great majority of our own working time went into producing those files correctly.

The reference rate is the number on the bedside monitor, and hospital regulation does not permit us to take it from the monitor's own output, so we photograph the screen with a camera of its own and read the digits back out of the image. That one constraint produced more code than any other part of the system. We began with a fixed homography, which drifted out of registration inside a single session (§6.2), and the chain of §4.3 is what that failure and the ones after it turned into: contour detection to locate the screen quadrilateral, KLT tracking to hold the corners through the moments when something passes in front of the screen, RANSAC to reject a corner that a reflection has pulled off the true quadrilateral, and a Kalman filter to smooth the corner trajectories without fighting a genuine nudge of the mount. Only at the end of that is there a rectified image worth reading digits from. We then stopped trusting the reader as well, which is where the three per-frame checks of §4.3.1 come from: one for a missing value, one for a value the voter has frozen at its last successful read, and one for a number no neonatal monitor would display. Over the whole cohort that leaves 98.487 % of 300,000 labeled frames usable and masks 1.513 % (§5.2). Establishing that the surviving labels sit against the right frames took another script: it compares change-points in the monitor's own rendered digit crop against change-points in the stored label array, which is how we know the label trails the display by 15 frames. Half a second is small against a 10 s window, and it only became small once the output had become a window.

Magnification is third-party code that we barely modified, and it was still weeks of work. The network is TensorFlow 1.x and will not share a Python environment with the PyTorch package the rest of the system uses, so it runs from its own virtual environment and hands over by writing folders of PNGs; every clip in the cohort makes that round trip once, at amplification 8 with a Butterworth pass-band of 0.2 to 2.0 Hz. We added one filter type to it, a butterLowpass that keeps the DC component of the amplified shape signal, which the band-pass variants throw away. The verification came afterwards, and it came in two parts, because the first part answered a weaker question than we had thought it did. Comparing the frames the trainer reads against the frames the magnifier was given tells us the two differ, which rules out the embarrassing case of having trained on raw video all along and says nothing about whether the magnification helped. The selectivity check exists because of that gap: per clip, the gain inside the pass-band against the gain above it, and on 16 of the 61 clips sampled the out-of-band gain is the larger of the two (§4.4). We report that figure as it came out.

Assembling the cohort was clerical work with one trap in it. Nine dated sessions hold 601 video directories between them (§4.2), and the convention that governs them is that a recording date is an infant: Patient_0 on two different dates is two different babies, and one baby has two records only where a single patient folder contains both Record_0 and Record_1. We got that wrong once, in the direction that matters. One recording had been entered under two subject identifiers, so the same infant sat on both sides of every leave-one-subject-out fold, and it went unnoticed until an audit in August, by which point every accuracy figure we had was inflated by it and had to be withdrawn (§6.4). The cohort file the system now reads, data/metadata_9inf_dedup.csv, holds 500 clips from 9 infants; 496 of them reach training, and the other four lose every one of their windows to the rule that a window needs at least half its frames labeled.

From there the preparation is arithmetic applied 5,431 times. Each 20 s clip gives 11 windows of 10 s at a 1 s hop, and each window is reduced on the 192 by 192 torso crop to 336 block-mean series at block sizes of 12, 24 and 48 px. Those series then have a linear and a quadratic basis projected out, get a Hann window and an FFT zero-padded to sixteen times the window length, are whitened by an f^-alpha law fitted per block over 5 to 280 bpm, resampled onto 160 logarithmic bins from 10 to 200 bpm, and collapsed to three order statistics across the pool; the cached ResNet features supply the other three channels of the (6, 160) array by the same route. Two clocks have to stay in step while that happens, because the pixel frames arrive at 30 fps and the cached features at 10, so one window is 300 samples on one side and 100 on the other, and its label is the masked mean of the reference over exactly the frames of the pixel window. The result is written once, to a single npz file. That is the decision that made the rest of the project affordable. Caching costs 3.48 s per clip, about 29 min for a pass over the whole cohort; training all 27 fold-seed models from what the pass produced takes 5.7 min.

A chain this long has to be checked stage by stage, because a defect near the front of it is invisible at the back. A stale monitor reading is still a plausible respiratory rate. A duplicated subject is a well-formed row in a CSV, and a magnifier that has amplified the wrong frequencies writes exactly as many PNGs as one that has not. Nothing crashes. The pipeline runs to the end and prints a number, and in the duplicate's case the number was better than the truth, which is the hardest sort of defect to go looking for. So each stage writes its output to disk and has a check of its own written against that output, and the reported figures are recomputed from the stored files instead of copied out of a log. Most of that checking code was written after the stage it checks was already in use, which is later than we would have liked.

5 Final Acceptance Tests and Results

Six of the seven tests below report measured results; the seventh, temperature, was not attempted. Every figure in this chapter is drawn from the stored per-clip prediction files, and every number in it was recomputed from those files and not read from a training log.

5.1 Measurement procedure and reported metrics

All measurements are made on the Jetson AGX Orin that also performs acquisition. Every accuracy figure is recomputed from stored per-clip predictions, so the numbers in this chapter and the numbers a reader could regenerate are the same numbers.

Let e be the prediction minus the reference over the scored units. We report:

```
MAE    = mean |e|
RMSE   = sqrt( mean e^2 )
bias   = mean e
r      = Pearson correlation( reference, prediction )
LoA    = bias +/- 1.96 * SD(e)          Bland-Altman limits of agreement
CI     = bootstrap, resampling INFANTS, not clips
```

- **Two granularities.** A window estimate covers 10 s. A clip estimate is the mean of the 11 window estimates within a 20 s clip, and is the quantity a bedside display would show. Unless stated otherwise, MAE means clip-level MAE.
- **Why validation is leave-one-subject-out.** A clip-level split would place clips of the same infant on both sides of the boundary. Kapoor and Narayanan catalogue this as a leakage type [34]: in the extreme, and unfortunately common, case the train and test samples come from the same people or units. The cost is measured. In longitudinal brain MRI, Rumala reports 97.33 % accuracy under record-wise cross-validation against 38.71 % on unseen subjects, an inflation of 59 points that also inverts the model ranking [35]; Saeb and colleagues argue from the same contrast that a clinical model must be validated the way it will be used [36]. Every accuracy figure here is measured on an infant the model has never seen.
- **Why the bootstrap is clustered on infants.** Eleven windows from one baby, or fifty clips from one baby, are not fifty independent observations. Resampling clips would produce a confidence interval several times too narrow and would make every difference look significant. Resampling infants is what makes the interval honest, and it is why the interval on 9.73 is as wide as [6.9, 12.9] with 496 clips. Reporting an interval at all is less routine than it should be: the NeurIPS 2019 reproducibility study found that 36 % of submissions declared error bars not applicable [33]. Every interval in this chapter is 20,000 clustered resamples, so the figures are quoted to one decimal place: the second decimal moves with the resampling seed and the first does not.
- **Why a no-skill baseline appears beside every MAE.** Respiratory rate in this cohort is widely spread, which is why a predictor that has learned nothing except the cohort median still incurs an MAE of 13.6 to 14.0 bpm. That number is what makes 9.73 interpretable: without it, a reader cannot tell whether the model has learned anything at all.
- **The reference is widely spread, and that sets the baseline.** Over the 496 scored clips the reference rate has mean 52.7 bpm, standard deviation 17.1, median 46.7, a 5th-to-95th percentile range of 31.4 to 84.3 and a full range of 19.5 to 97.5. A third of the cohort, 34.9 %, sits outside the 30 to 60 bpm band usually quoted for quiet neonatal breathing, almost all above it. The no-skill MAE of 14.02 bpm against the median, 14.58 against the mean, is therefore high because the cohort is spread and not because the reference is noisy.
- **Two-sided reporting of within-subject performance.** An overall correlation can be produced entirely by between-infant differences in mean rate, without the model tracking anything within a

single baby. We therefore report the within-subject correlation separately, and it is the more demanding of the two figures.

5.2 Ground-truth label quality

Figure 5 and Table 10 give the measurement over the whole nine-infant cohort: all 500 clips, 600 frames each, 300,000 labeled frames, using the label QC of §4.3.1. The per-defect counters are the ones emitted into the metadata CSV by `build_metadata_from_magnified.py`, and the three of them are disjoint by construction, because each test is applied only to frames that survived the previous one. That is what lets them be added: the three counters sum to 4,538 frames and 1 minus the mean valid fraction over the 500 clips is 1.513 %, the same number to four decimal places. The out-of-range run statistics quoted in §4.3.1 are counted over the raw monitor series before any clip is rejected, so they are not comparable to these.

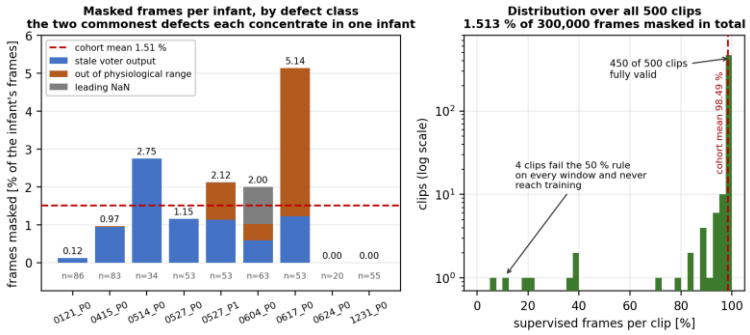


Figure 5: *Measured ground-truth label quality over the whole nine-infant cohort. Left: masked frames per infant, split by defect class. Neither of the two commonest defects is spread evenly: 371 of the 377 NaN frames belong to* `0604_P0` *and 1,243 of the 1,731 out-of-range frames belong to* `0617_P0`*, which is also the infant with the anomalous label lag of §4.3.2. Two infants are perfectly clean. Right: the distribution over all 500 clips on a logarithmic count axis, showing the 450 fully valid clips and the small tail of badly supervised ones, four of which fail the half-labelled rule on every window and never reach training.*

Table 10: *Measured label quality, all nine infants, 500 clips, 300,000 frames.*

Quantity	Frames	Share	Concentration
Total labeled frames	300,000	100.000 %	
Supervised (valid) frames	295,462	98.487 %	
Masked: stale voter output	2,430	0.810 %	Present in seven of nine infants
Masked: out of physiological range	1,731	0.577 %	1,243 of them in `0617_P0`
Masked: leading NaN	377	0.126 %	371 of them in `0604_P0`
Masked, total	4,538	1.513 %	0.00 % for two infants, 5.14 % for the worst
Clips fully valid	450 of 500	90.0 %	
Clips that never reach training	4 of 500	0.8 %	Every window fails the half-labelled rule

The defects are not cohort-wide but concentrated: 371 of the 377 NaN frames are in `0604_P0`, 1,243 of the 1,731 out-of-range frames in `0617_P0`, and that one infant carries 36 % of all masked frames. And 450 of 500 clips are entirely clean, so masking costs coverage on a minority of clips instead of a little on all of them.

5.3 Failure of the dense transformer design

This is a negative result and the most informative measurement in the project. The design of Figure 6 was implemented in full and trained. Its held-out predictions were nearly constant within each infant while differing between infants, the signature of a model that has learned to recognize the patient instead of measuring the patient. Geirhos and colleagues name this shortcut learning, with the rule that one should never attribute to high-level abilities that which can be adequately explained by shortcut learning [37].

Table 11: *Measured diagnosis of the dense transformer design.*

Measurement	Result	What it establishes
Infant identity from pooled features	A classifier on the pooled 2048-D ResNet features identifies which infant is in the clip with 96.3 % accuracy	The representation carries identity almost perfectly, so identity is available to be exploited
Held-out prediction spread within an infant	Collapsed to 0.3 to 2.1 bpm	The model emits one number per infant. It is not tracking breathing at all
Selectivity of a single pooled measure	On a 12 px grid the best-scoring block beats the whole-region mean on 22 of 27 clips, and where only the whole-region mean is available the rate is not recoverable at all on 8 of those 27	A scalar averaged over the whole region is not selective enough. Breathing occupies a small part of the frame, so an average over everything else dilutes it
Spectral energy at the true rate	Averaged across all 2048 feature channels, the energy at the reference rate is lower than for raw pixel blocks on 25 of 27 sampled clips. Measured per channel it is not: the strongest individual channels reach a signal-to-noise ratio of about 172 at the true rate, and on one infant the feature peak matches the label to 0.44 bpm, better than raw pixels at 0.58	The encoder does not discard breathing. Any statistic averaged across the 2048 channels dilutes it, because the rate lives in a handful of them. This is the measurement that fixed the front end: the delivered channels are a maximum and a 90th percentile as well as a mean, and the first two exist precisely to survive this dilution

The third row is the mechanism, and it was asserted before it was measured, so we measured it: the same 27 clips, block time series on the torso crop at five granularities from a 16 by 16 grid down to a single cell, scoring the peak at the reference rate. A single cell is the whole-region average, which is what the first architecture computed per frame. Two columns of Table 12 need definitions, because the obvious reading is wrong. The best-block score is a maximum over however many blocks the grid has, 256 at 12 px against one at 192 px, so part of its decline is an order-statistic effect and at 192 px it equals the median-block column exactly. The median-across-blocks column moves the other way, and we print it instead of hiding it: most fine blocks contain no chest, so their median is low by construction.

The best 12 px block beats the whole-region mean on 22 of the 27 clips individually, which selection alone would not deliver so lopsidedly. And rate error is a different quantity from score: the fine grid places its peak within a median of 5.5 bpm of the reference, while the whole-region mean manages 12.5 bpm on the 19 clips where it produces a peak at all, mean error 20.7, and none on the remaining 8. Excluding those 8 makes the 12.5 an understatement.

So the conclusion is narrower than the one we first drew, and it is the one the delivered front end acts on. Averaging over a region does not cancel a respiratory signal; it dilutes it, along with everything else in the region, until a competing periodic source anywhere in the frame can outweigh it. The remedy is to pool with an order statistic instead of a mean, and to pool last instead of first (§4.5).

Table 12: *Spatial pooling against signal. 27 clips drawn from the pre-deduplication metadata, so eight of the nine infants are represented. The last row is the whole-region average.*

Spatial granularity	Block size	SNR of the best block	SNR of the median block	Median rate error	Clips giving a peak
16 x 16 = 256 blocks	12 px	1.273	1.011	5.5 bpm	27 of 27
8 x 8 = 64 blocks	24 px	1.274	1.026	11.5 bpm	27 of 27
4 x 4 = 16 blocks	48 px	1.244	1.044	11.5 bpm	27 of 27
2 x 2 = 4 blocks	96 px	1.182	1.061	10.5 bpm	26 of 27
1 x 1, whole-region average	192 px	1.087	1.087	12.5 bpm	19 of 27

We first tried to repair the pooling in place. Two alternatives to the spatial mean were scored on the same 27 clips, pooling by energy in the respiratory band and a learned attention pool, with the plain average as control: median error at the reference rate 22 bpm for the average, 19 for band energy, 19 for attention, with 5, 6 and 5 clips of 27 within 3 bpm. Neither recovers the rate, and that is what made the redesign necessary: the problem is that pooling happens before the frequency analysis, not which pooling function is used. The code is kept under `tools/attempted_pooling_fix/`. This paragraph and Table 12 measure the same control with different instruments, one voting over blocks and one reading a pooled spectrum, which is why the whole-region figure is 12.5 bpm there and 22 here.

The dilution and the dataset compound. Spatial pooling leaves the stable, infant-specific appearance intact, so the optimizer was offered a representation in which identity was easy and breathing was almost absent, together with a dataset in which knowing the infant predicts the rate reasonably well. It did the sensible thing.

This failure mode is documented in medical imaging and is not peculiar to this project. Zech and colleagues found a pneumonia classifier that identified the acquisition hospital from a chest radiograph on up to 99.98 % of test images [38]; Badgeley and colleagues report hip-fracture detection falling from an AUC of 0.91 to chance once patient and process variables were matched out [39]; Oakden-Rayner and colleagues give the general form [40]. Chaibub Neto and colleagues name our exact case, where the confound is the subject and not the site, identity confounding [41].

We could not have found this earlier. Our first validation setup was a single fixed split inside one infant's recordings, and with one identity in the data there is no identity memorization to see. It became visible only once we had nine infants and moved to leave-one-subject-out, which is also why the audit in §6.4 matters.

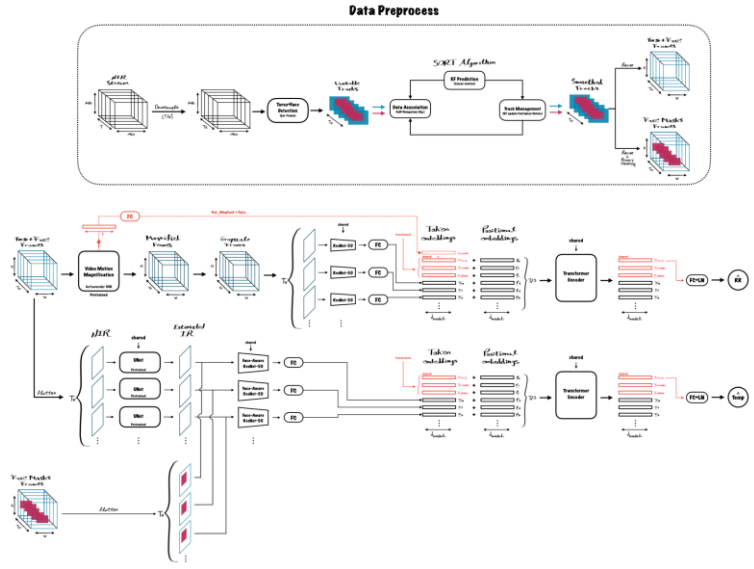


Figure 6: *The architecture specified in the preliminary report and built in the first half of the year (original diagram by Maor Oz). The respiratory-rate branch, from the magnified frames through the shared transformer to the dense output, is the path measured in §5.3. Nothing in that diagnosis is specific to this design's authorship: any architecture that pools spatial features globally and is trained on nine infants offers the same shortcut, and it was the measurement instead of the literature that made it visible.*

5.4 Respiratory-rate accuracy under leave-one-subject-out

This is the acceptance test for the product's declared purpose, and Figure 7 and Table 13 give the result. Nine folds by three seeds gives 27 models; the seed distributions are averaged before the rate is read out, and every infant serves as the held-out test subject exactly once.

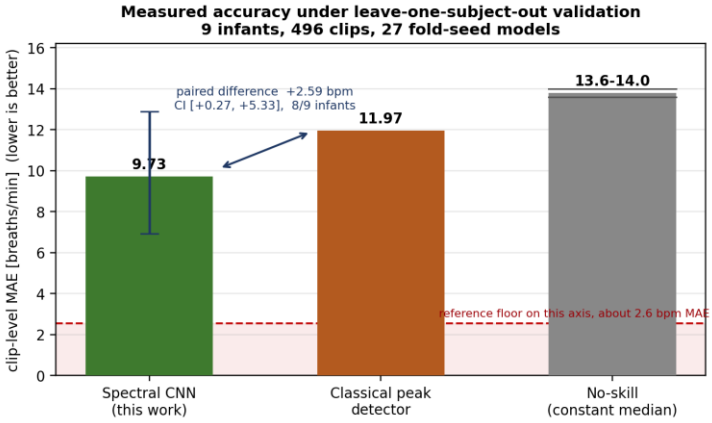


Figure 7: *Measured clip-level accuracy, recomputed from the 496 stored per-clip predictions. Every interval is a bootstrap clustered on infants, which is why they are as wide as they are with 496 clips. The shaded band is the floor the reference imposes on this axis, about 2.6 bpm of mean absolute error.*

Table 13: *Measured respiratory-rate accuracy, nine-infant cohort, leave-one-subject-out.*

Quantity	Value	Reading
Clip MAE, delivered system	9.73 bpm, 95 % CI [6.9, 12.9]	The headline result. Nested early stopping, `runs/31_windowed`
Clip MAE, fixed 150 epochs	9.26 bpm, 95 % CI [6.6, 12.4]	The same architecture with the stopping rule removed, `runs/40_fixed150_loso`. Reproduced at 9.25 bpm by an independent repeat, `runs/41_curves150`
Clip MAE, a second nested-stopping run	9.57 bpm, between-seed SD 0.14	`runs/33_final`, the same protocol with different seeding. So the nested-stopping figure is 9.6 to 9.7 across two runs, which is the reproducibility of the number
No-skill baseline, cohort median	14.02 bpm	A predictor that has learned only the middle of the cohort. The system carries information
Oracle baseline, each infant's own mean rate	7.26 bpm	Not admissible as a baseline, and reported because every label-informed figure is declared as one. It needs labels from the held-out infant, which the protocol forbids. It bounds what within-infant tracking could be worth
Classical peak detector, honest per-fold calibration	12.31 bpm	The detector calibrated within each fold, never on the test infant, and scored against the same label as every other row of this table. It measures 11.97 against the whole-clip label instead
Paired advantage of the CNN over the detector	+2.58 bpm, CI [+0.4, +5.2]; the CNN is better on 8 of 9 infants	Significant, and fragile. Two infants each carry it on their own
Overall Pearson r	0.69	Includes between-infant variation
Within-subject Pearson r	0.288, CI [0.123, 0.409]	Excludes zero, so part of the signal is within-infant and not only between infants. r = 0.288 is weak, about 8 % of the within-infant variance, and the interval maps to 1.5 % to 17 %. A model that had learned only each baby's average would give zero
Reference noise floor	3.2 bpm	The monitor's own within-window SD
Window-level MAE, for completeness	10.93 bpm	Before the 11 windows of a clip are averaged. The averaging is worth 1.2 bpm

The classical detector emits a peak sharpness per clip, and that sharpness needs no label, so it can gate the output. Gated at the median sharpness the detector scores 7.42 bpm on half the clips; at the sharpest 30 % it scores 3.66 bpm against a no-skill baseline of 14.89 on that same subset, with 8 of 9 infants still represented. A bedside device could report a rate when the spectrum is sharp and stay silent otherwise.

Three further measurements bound the headline number. A single constant per infant, chosen leave-one-clip-out from that infant's own other clips, scores 7.26 bpm against our 9.73; it is an oracle and not a competitor, because it needs labels from the infant being tested, but it bounds what is left to gain, and adding an oracle per-infant offset to our own predictions gives 7.21, so almost all of our error is in the

level. Second, the detector advantage is fragile: jackknifed over infants it keeps an interval excluding zero on seven of the nine leave-one-out subsets, but dropping `1231_P0` gives [-0.07, +3.01] and dropping `0617_P0` gives [-0.01, +5.18]. It is real on the cohort as measured and it is carried by two infants. Third, splitting the cohort into thirds by how much the reference moves inside a clip, the learned read-out degrades from 7.00 to 11.35 bpm while the detector degrades from 5.92 to 17.39, and the paired advantage is significant only on the most variable third, +4.90 bpm with an infant-clustered interval of [+2.05, +7.51]. The pooled means suggest the detector is better on the steadiest third, 5.92 against 7.00, but computed per infant that reverses sign with an interval of [-3.0, +10.1], so the crossover is not established and we do not claim it. After those deductions the defensible claim is that the detector cannot be substituted for the learned read-out on the hardest third of the data.

One property of the training removes a degree of freedom an examiner would otherwise be right to question. Validation loss rises by 24 % after epoch 5, which is textbook overfitting, and it does not matter: the quantity read out is the expectation of a distribution and not the loss, so the distribution can sharpen wrongly while its centre of mass stays put. Clip MAE is flat from epoch 11 to epoch 150, and the entire excursion in validation MAE across 150 epochs is 1.5 bpm, well inside the resolvable 2.17. That is why the fixed-150 figure of 9.26 can be quoted beside the early-stopped 9.73 without the stopping rule carrying the result, and it is why one fold selecting epoch 0 (`0514_P0`, seed 1) does not invalidate it. Three checkpoint files exist and one is misstamped: `runs/35_final_150ep/model.pt` carries 150-epoch weights labelled with the 9.57 bpm belonging to the nested-stopping protocol of `runs/33_final`, so cite `runs/35_final_150ep/model_measured.pt` for the 150-epoch configuration.

5.5 Structure of the error

We report agreement in the Bland-Altman form [42], which plots the difference between two measurements against their mean and gives limits of agreement because two methods can correlate strongly and still disagree widely.

Table 14: *Measured error structure.*

Quantity	Value	Interpretation
Bland-Altman bias	-0.86 bpm	Almost unbiased. The system is not systematically high or low
Limits of agreement	-25.4 to +23.6 bpm	Wide. Individual estimates cannot be trusted clinically
Limits of agreement, repeated-measures form	-26.1 to +24.4 bpm	The form that accounts for several clips coming from one infant. The method as originally published [42] assumes one observation per subject; the adjustment for repeated measures is a later refinement by the same authors. It is the correct form here and it is 3 % wider, so the naive limits are not flattering us by much
95 % interval on the lower limit	-33.6 to -15.2 bpm	The limits are themselves imprecise at nine infants. Bland and Altman's analytic standard error would give plus or minus 1.9 bpm here; clustering the bootstrap on infants gives plus or minus 9. Reporting the analytic version would understate the uncertainty five-fold
95 % interval on the upper limit	+17.9 to +28.2 bpm	As above
Proportional bias	Difference regresses on mean with slope -0.63, CI [-0.87, -0.41]	Excludes zero, so a single pair of limits does not hold across the range. This is the compression below seen on the Bland-Altman axes, not a second defect
Distribution of the differences	Skew -0.23, excess kurtosis +0.26, 95.0 % inside the limits	The normality the limits assume holds, so the construction is valid
Clips within 5 bpm of the reference	37.1 %	A useful, blunt statement of practical accuracy
Windows within 5 bpm, before averaging	34.2 %	The model's native output. Averaging the 11 windows of a clip is worth 2.9 points here, and takes the MAE from 10.93 to 9.73 and the 90th percentile of the error from 23.6 to 20.3 bpm

Clip RMSE	12.52 bpm	The metric the declared target was written in. An MSE target has to be compared against RMSE, not MAE
Clip MSE	157 bpm^2	Against a declared target of at most 2.0 bpm^2
Slope of predicted on reference	0.40	Predictions are compressed toward the cohort mean

Two of those rows are checks the method requires. The limits carry their own uncertainty, large at nine clusters: the lower limit could honestly be anywhere from -34 to -15 bpm. And one pair of limits does not describe the whole range, because the difference falls with the mean at a slope of -0.63. Both follow from the compression the slope of 0.40 already reports, so neither is new information; what they change is how the interval should be quoted.

The RMSE is recomputed from the 496 stored per-clip errors and also follows from the printed limits, which is a consistency check: SD = (23.6 + 25.4) / (2 x 1.96) = 12.50 bpm and RMSE = sqrt(0.86^2 + 12.50^2) = 12.52. The MSE is 157 bpm^2, reported because the target was declared as an MSE and because comparing MAE against it understates the shortfall by a third. For the fixed-150 configuration the same quantities are 12.29 bpm and 151 bpm^2.

The compression is the defect to attack, and three mechanisms can each produce a slope below one with a near-zero bias.

- **Attenuation by label noise.** Error in the reference regresses the fitted slope toward zero. The reference has a within-window standard deviation of 3.2 bpm (§4.3.2), so this term is bounded, and bounding it properly needs the between-clip spread of the reference, which we have not reported.
- **Shrinkage in the read-out.** The estimate is the expectation of a softmax over a bounded logarithmic axis whose center lies near 45 bpm. Any broad output distribution is therefore pulled toward mid-range, and a slope below one with almost no bias is exactly that signature. This is a property of our own read-out, not of the signal, and it was testable without retraining, by recomputing the rate as the interpolated argmax of the same stored distributions instead of their expectation. That test has now been run and Table 15 gives it.
- **Weak evidence reaching the read-out.** Region placement, the magnification amplification factor and pass-band, and the window length all set how much respiratory energy survives into the spectrum. This is the term the recommendations in §7.7 address.

The first can be bounded arithmetically. Reference error regresses a fitted slope toward zero by one minus the ratio of reference variance to reference spread; the no-skill MAE of 13.6 to 14.0 bpm implies a between-clip reference SD near 17 bpm, so with 3.2 bpm of reference error that factor is about 0.97. Label noise accounts for roughly 0.03 of the 0.60 shortfall and is not the explanation.

The second was tested without retraining: the 5,431 stored distributions re-read three ways, as the expectation the system uses, as an interpolated argmax, and as a median. On synthetic distributions carrying a correct narrow peak on a 55 % uniform floor, where every departure from unit slope must be the estimator's own shrinkage, the expectation returns 0.365 and the argmax recovers 1.000 exactly, so the test can detect shrinkage when that is what is happening.

Table 15: *Three read-outs of the same 5,431 stored distributions, scored at clip level. No model was retrained. Intervals are bootstrap resampled over infants, 10,000 draws.*

Read-out	Clip MAE	Paired difference against expectation	Slope	Slope 95 % CI	r
Expectation (delivered)	9.26	reference	0.448	[0.21, 0.62]	+0.704
Interpolated argmax	9.24	-0.02, CI [-2.3, +1.9]	0.538	[0.26, 0.72]	+0.684
Distribution median	8.48	-0.78, CI [-1.9, +0.1]	0.531	[0.24, 0.73]	+0.726

On the real distributions the slope moves only from 0.448 to 0.538, a rise of 0.090 against the 0.552 separating 0.448 from unity, and clip MAE does not move, 9.26 to 9.24. The three slope intervals overlap almost completely, so the rise is not itself resolvable, but all three exclude 1.0. No choice of read-out over these distributions recovers the reference range, so shrinkage is not the dominant term and the compression is a property of the evidence reaching the read-out. Mean peak mass is 0.084 and mean entropy 3.64 nats against 5.08 for uniform: the outputs are broad but not flat, carrying real information, r = +0.704, that is not sharp enough to place the rate.

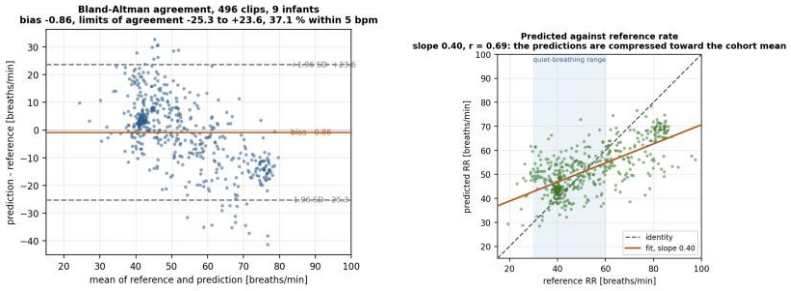


Figure 8: *Left: Bland-Altman agreement over all 496 clips. The bias is negligible at -0.86 bpm while the limits of agreement run from -25.3 to +23.6 bpm, so an individual estimate cannot be trusted clinically even though the method is unbiased on average. Right: predicted against reference rate, with the identity line and the fitted line at slope 0.40. The cloud tilts the right way and far too little, which is the compression discussed below. Both are drawn from the stored per-clip predictions.*

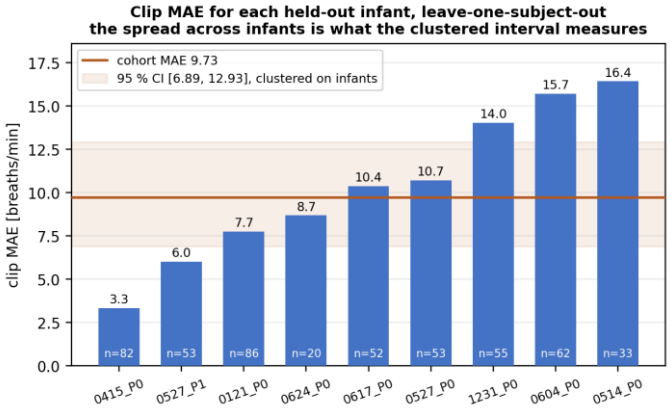


Figure 9: *Clip MAE for each held-out infant, sorted. The best fold is 3.3 bpm and the worst 16.4 bpm, a five-fold spread across nine infants, and that spread is exactly what the infant-clustered interval on 9.73 measures. Fold size does not explain it: the worst fold has 34 clips but the second worst has 63, the second largest of the nine, and the smallest fold at 20 clips scores 8.69, better than the cohort mean. Nor does label quality, since the most heavily masked infant sits mid-table at 10.37. What the three worst folds share is a high mean rate, 61, 65 and 82 bpm against a cohort mean of 53, which is the compression of §5.4 sorted by infant.*

5.5.1 Testing for collapse to a per-infant constant

Table 16 compares, for each held-out infant, the standard deviation of the predictions against that of the reference over the same clips. A model that has memorized one rate per infant produces the first column near zero whatever the second does.

Table 16: *Held-out prediction spread within each infant, against the reference spread.*

Held-out infant	Clips	SD of predictions	SD of reference	Prediction range
0121_P0	86	6.69	11.46	44.4 to 78.3
0415_P0	82	1.51	1.04	40.1 to 48.9
0514_P0	33	5.84	18.27	48.7 to 74.4
0527_P0	53	9.30	12.68	28.5 to 68.5
0527_P1	53	5.21	6.36	32.1 to 54.5
0604_P0	62	8.40	16.39	35.4 to 69.3
0617_P0	52	6.43	7.35	30.8 to 66.3
0624_P0	20	2.73	7.40	42.0 to 55.0

1231_P0	55	3.14	3.19	54.7 to 74.7
Pooled over the cohort	496	6.09	10.45	
The design this replaced		0.3 to 2.1	6.5 to 16.6	

Table 17: *Within-subject correlation for each held-out infant. The pooled figure is 0.288.*

Infant	0121_P0	1231_P0	0604_P0	0527_P0	0617_P0	0415_P0	0527_P1	0514_P0	0624_P0
Within-subject r	+0.53	+0.50	+0.35	+0.29	+0.13	+0.11	+0.09	+0.03	-0.28

The pooled 0.288 is therefore an average over a real spread: on two infants the system tracks the rate usefully, on one twenty-clip fold it moves against the reference, and on the rest it carries a weak positive signal. Whether the three that track share anything, in pose, blanket state or reference stability, is a question three positives cannot answer.

The collapse is gone. Prediction spread within an infant runs from 1.5 to 9.3 bpm against 0.3 to 2.1 for the design of §5.3, pooled 6.09 against a reference spread of 10.45. That ratio needs care, because the flattering reading is wrong: a spread ratio of 58 % is not 58 % of variation explained, and noise of the right magnitude would score the same. The variation actually explained is the square of the within-subject correlation, 0.288 squared, which is 8.3 %. The corresponding slope is 0.168 within an infant, against the 0.40 of §5.5 which is inflated by differences between infants.

Two rows need care. `0415_P0` has a prediction spread of 1.51 bpm, which looks like collapse until you notice its reference spread is 1.04: that infant breathed at a nearly constant rate, so a near-constant prediction is correct, which is also why it is the best fold at 3.3 bpm. It also means the lower end of the 1.5-to-9.3 range is set by an infant who did not vary, so the contrast with 0.3 to 2.1 rests on the other eight. `1231_P0` is the real weakness: reference spread 3.19, prediction spread 3.14, yet MAE 14.03, because its mean rate of 82 bpm is the highest in the cohort and the predictions sit 14 bpm below it. That fold fails on level, not spread.

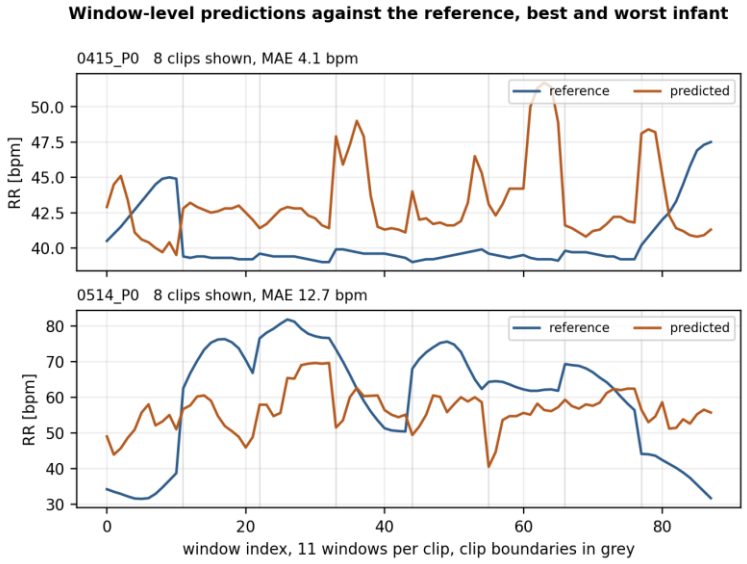


Figure 10: *Window-level predictions against the reference for the best and the worst fold, clip boundaries in grey. This is the output a bedside display would show, one estimate per second. The MAE printed on each panel is over the eight clips drawn, so it differs from the whole-fold figures of 3.3 and 16.4 bpm in Figure 9. The upper panel is what the system does when it works: the trace follows the reference within a couple of breaths per minute and holds across clip boundaries, which matters because each clip is scored independently. The lower panel is the worst fold, where the predictions carry roughly the right shape at the wrong level. Neither panel shows a flat line, which is what the first architecture produced.*

The delivered package includes that view as two rendered videos. Each plays a 20 s clip with the magnified torso region beside the original near-infrared crop, so amplified chest motion can be seen

against the unamplified source, and a trace underneath advances one estimate per second against the step-wise monitor reference. They are `0527_P1_Patient_1_Record_0_vid_27`, predicted 31.8 against 31.4 bpm, and `0415_P0_Patient_0_Record_0_vid_6`, predicted 41.4 against 39.5, both from the fixed-150 configuration; §5.4 reports accuracy over the whole cohort. The two panels show the magnified video against the original video, not one prediction against another: what the magnification stage is worth to accuracy is the ablation §7.7 recommends, and it has not been run.

5.5.2 Identity is still present in the input

§4.6 argued that the redesign removes the route by which appearance reached the output, and §5.5.1 shows the symptom is gone. Neither answers the direct question, which is whether the information is still there. It is, and this subsection is the one place in the report where a measurement contradicted what we expected to find.

The probe is the nine-way logistic classifier of §5.3 with a different input: the clip-mean (6, 160) spectra the read-out actually consumes, on a split of 248 training against 248 test clips that is disjoint by clip. Two baselines matter. Uniform guessing scores 11.1 %, and always naming the largest infant scores 18.5 %.

Table 18: *Nine-way infant identification from the model's own input. The comparison row is the pooled ResNet features of the abandoned design, measured in §5.3.*

Input given to the classifier	Clip level	Window level
All six channels	85.5 %	76.8 %
Channels 1 to 3, the pixel-block spectra, only	85.1 %	73.8 %
Channels 4 to 6, the ResNet-derived spectra, only	71.8 %	59.5 %
Pooled ResNet features of the abandoned design (§5.3)	96.3 %	
Floor: always name the largest infant	18.5 %	18.6 %
Floor: uniform guessing over nine infants	11.1 %	11.1 %

A control is needed first. The nine infants sit in very different rate bands, `0415_P0` at 40.0 plus or minus 1.0 bpm against `1231_P0` at 82.0 plus or minus 3.2, so a spectrum peaked at the breathing rate names the infant through the label, which is the target and not a shortcut. The true reference rate alone identifies the infant on 39.1 % of clips and the six channels' own peak frequencies reach 45.6 %, so the rate accounts for some of the 85.5 % but not most. The decisive test holds the rate fixed: restricted to clips between 35 and 50 bpm, a band where five infants overlap, the rate alone scores 44.7 % against a 38.8 % floor while the full spectra score 93.2 %. The surviving decodability is appearance or recording-session information, and holding the rate fixed makes it easier to read, not harder.

Two things follow from that. First, a correction: §4.6 cannot claim the transform removes the infant's identity from what the read-out sees, and now states the narrower claim instead. The information is present and the read-out cannot route it to the output, which also predicts that a less constrained read-out over these spectra should re-acquire the shortcut. Second, the information is not where §4.6 guessed: channels 1 to 3 reach 85.1 % alone, so it is carried by the pixel-block spectra and any sentence blaming the ResNet-derived channels is wrong. One limit belongs with the probe: it is a linear model fitted with the labels on other clips of the same infants, an upper bound on availability and not a measurement of use.

The second half of the question is what the ResNet channels are for, and the answer is nothing measurable. Table 19 retrains the whole leave-one-subject-out protocol on channels 1 to 3 alone against the fixed-150-epoch six-channel baseline, on identical clips and identical labels.

Table 19: *Dropping the three ResNet-derived channels. Same protocol, same 496 clips, same labels. The paired difference is clustered on infants.*

Configuration	Clip MAE	95 % CI	Slope	r

Six channels, fixed 150 epochs	9.26	[6.7, 12.5]	0.448	+0.704
Channels 1 to 3 only, fixed 150 epochs	9.25	[6.2, 12.6]	0.467	+0.719
Paired difference	-0.01	[-1.5, +2.0]		
No-skill: mean of the other eight infants	15.7			

The difference is -0.01 bpm with its whole interval inside the 2.17 bpm this cohort resolves, so the frozen ResNet-50 can be removed at no measured cost. That is an engineering result: it deletes the feature-caching stage and the 875 MB `feats_9inf.npz` from the deployment path, and makes the magnification ablation of §7.7 cheaper and cleaner, since the pixel-block channels are the only place magnification can act. One caveat travels with it: the cohort null hides a per-infant reshuffle, six better, three worse, `1231_P0` worse by 6.5 bpm. No cohort-level difference is not the same statement as no difference.

5.6 Hypotheses that were tested and refuted

Six claims were made at some point during development and did not survive re-measurement on the clean cohort. They are reported here because a project that only lists its successes is not reporting its method.

Two numbers govern how any of this should be read, and they are the reason the list below is a list of retractions and not of results. Each infant contributes one paired difference in MAE between two systems; the standard deviation of those nine differences over the square root of nine is the standard error, and 1.96 times it is the smallest effect this cohort can resolve. On the detector comparison that is 2.17 bpm. Resolving 1 bpm would need the same spread over 43 infants. Every architectural comparison run during the project produced a difference smaller than 2.17 bpm, so those comparisons were not measurements: with nine infants we could not have detected whether a change helped even if it had.

One caveat belongs with the detector comparison, which is the single claim that does clear the bar. It is one of seven paired tests in this chapter, and a 95 % interval applied seven times excludes zero once by chance alone; its lower bound is +0.27 bpm, so a Bonferroni-corrected interval over seven comparisons would include zero. Two things keep it in the report. The detector was our designated baseline from the outset and was not one candidate picked out of several after the fact, and it is the only effect clearing the uncorrected threshold at all. It is therefore also the only architectural claim we make, and confirming it needs the larger cohort of §7.7 and not a different statistic.

Table 20: Refuted claims, with what each quantity actually is.

Claim, as once made	What re-measurement showed
A confidence score identifies unreliable clips: per-infant correlation +0.710, and the sharpest quartile reaches MAE 2.94 bpm	Per-infant correlation is +0.029, and the sharpest quartile is 8.18 bpm. The score carries almost no information about reliability. The docstring in the source has been corrected
Restricting the region of interest to the chest and torso improves accuracy	+0.79 bpm, CI [-0.62, +3.13], better on only 4 of 9 infants. Excluding one infant it is worse, 9.20 to 9.48. Not supported
The classical detector reaches MAE 9.03 bpm	That figure came from in-sample calibration on a gated subset that still contained the duplicate. Honest per-fold calibration gives 11.97 bpm
Both final checkpoints score 9.57 bpm	9.57 was earned by nested early stopping at median epoch 11 (`runs/33_final`). The fixed-150-epoch configuration is a different model and measures 9.26 bpm. The claim came from `runs/35_final_150ep/model.pt`, whose metadata block was copied forward without being re-measured; §5.4 gives the correct mapping
Blending the detector with the CNN improves on either	Fitted stack 10.12 bpm, 50/50 average 10.30 bpm. Both worse than the CNN alone
Any accuracy figure from the runs numbered 11 to 18, for example 7.82 bpm	All were computed with `legacy_P0` present and are void (§6.4)

5.7 Body temperature

Not attempted. The obstacles were in the data and not in the model. The declared target of MSE at most 0.2 °C is untested and no temperature figure is claimed anywhere in this report. The bedside monitor displayed a temperature reading for too few of the recorded infants to assemble a reference we would trust, and where it was displayed it was sometimes on a separate monitor that would have needed another camera, so there was no ground truth to train against on most of the cohort. The input side was not in place either: the near-infrared to infrared translation network [43] written for this branch needs registered NIR and IR frame pairs, and no such pairs were recorded.

6 Problems Encountered and How They Were Solved

The procedure asks for the significant problems of planning, implementation and operation, and how they were overcome. We group them by where they arose. Each entry gives the symptom, the diagnosis and the resolution, and the diagnosis is the part with engineering content in it, because in most of these cases the symptom pointed somewhere other than the cause.

6.1 Acquisition and hardware

1. **The purchased cameras did not support the flashed OS image.** A hardware and software compatibility mismatch between the camera SDK and the Jetson image. We identified the JetPack version supporting both the camera SDK and the AGX Orin and re-flashed the unit.
2. **The internal storage filled within days of recording.** Real video volumes are nothing like test clips. A 1 TB NVMe SSD was added, which the budget had not carried (§7.5).
3. **A Python and OpenCV capture loop dropped roughly one frame in five.** Frame loss breaks the common time base every downstream stage assumes. Replaced with a GStreamer [44] pipeline, which records all streams at 30 fps with no loss.
4. **No commercial mount met the clinical constraints.** The rig has to hold its pose between sessions, leave the working volume free, and not vibrate, because a vibrating mount produces exactly the periodic displacement stage 3 amplifies. We designed and 3D-printed articulating arms and connectors.
5. **The ward acquisition service and the training job compete for the same machine.** The service outranks the pipeline and must never be killed or renamed to free resources. The training job is deprioritized with `nice` and `ionice` instead, which is why the timings of §3.5 are quoted as measured on a shared machine.

6.2 Ground-truth extraction

1. **The monitor is photographed obliquely, and both it and the camera move slightly.** Hospital regulation ruled out reading the values from the monitor's own data output, so they had to be recovered from a photograph of its screen. A fixed homography would drift out of registration within a session. We built a tracking chain instead: screen-quad detection, KLT feature tracking, RANSAC homography estimation and Kalman-filtered corner smoothing with outlier rejection, then perspective rectification and digit reading (§4.3).
2. **The recovered digits contain three distinct classes of defect.** Missing readings, readings frozen at a stale value, and readings outside any physiological range. Each is detected by its own test and the affected frames are masked and not interpolated; §5.2 reports 1.513 % of 300,000 frames masked cohort-wide, and a window is dropped if fewer than half its frames survive.
3. **The stored label does not correspond to the frame it is stored against.** Assumed negligible in the preliminary report. Measured in §4.3.2 at -15 plus or minus 1 frames, half a second, constant on eight of nine infants. Small against a 10 s window and fatal to the per-frame output originally planned.

6.3 The architecture

1. **Held-out predictions were nearly constant within each infant while differing between infants.** The model was solving an easier problem than the one posed: pooled ResNet features identify the infant with 96.3 % accuracy, and within-infant prediction spread had collapsed to 0.3 to 2.1 bpm. Diagnostics `step_a` through `step_h` isolated the mechanism instead of tuning the hyper-parameters (§5.3).
2. **The respiratory signal was absent from the features the model was given.** Not cancelled but diluted: breathing occupies a small part of the frame, so a mean over the region buries it, and §5.3

measures the best 12 px block beating the whole-region mean on 22 of 27 clips. ResNet features also carry less energy at the true rate than raw pixels on 25 of 27 clips. We replaced the spatial mean with order statistics over pixel blocks at three scales and demoted the ResNet features to three of six spectral channels.

3. **Any representation retaining appearance offers the identity shortcut.** Suppressing it by regularization would have been a matter of degree, and nine infants is not enough data to police it. We removed appearance from the representation instead: a power spectrum on a logarithmic rate axis holds no spatial information, and a shift-equivariant read-out cannot encode absolute position, so the measured shortcut is unrepresentable and not merely penalized. §5.5.2 measures the limit of that, since the identity information is still decodable from the input at 85.5 %.
4. **A point regression on bpm fits a reference that is itself uncertain.** The label has a 3.2 bpm noise floor and breathing spectra have harmonics that a point target penalizes arbitrarily. We replaced the Huber loss with soft-target cross-entropy over frequency bins, a Gaussian centred on the reference rate.
5. **Frozen BatchNorm silently corrupted the frozen backbone.** `requires_grad = False` freezes weights but not BatchNorm running statistics, and `nn.Module.train()` recurses into every submodule, so an ordinary `model.train()` puts the backbone back into batch-statistics mode. The hazard is documented [45]. We overrode `train()` so the trunk stays in `eval()` at all times, and the fix is retained in the cached-feature path of the delivered system.

6.4 Validation and measurement

1. **One infant was present twice under two identifiers.** `legacy_P0` was the same recording as `0604_P0`, so it appeared on both sides of leave-one-subject-out folds and inflated every result obtained before 23 August 2026, including a 7.82 bpm figure. The cohort was deduplicated into `data/metadata_9inf_dedup.csv` and every affected number was withdrawn (§5.6).
2. **A clip-level split leaks the subject.** Clips of one infant on both sides of the boundary make the test set a memory test. Replaced with nested leave-one-subject-out, every infant held out exactly once (§5.1).
3. **A bootstrap over clips makes every difference look significant.** Fifty clips from one baby are not fifty independent observations. Every interval in this report resamples infants, which is why the interval on 9.73 is as wide as [6.9, 12.9] with 496 clips.
4. **A checkpoint carried metrics it had not earned.** `runs/35_final_150ep/model.pt` was stamped with the 9.57 belonging to the nested-stopping protocol. The report cites `model_measured.pt` and says so in print (§5.4).
5. **Several claims did not survive re-measurement.** Six are listed in §5.6 with what each actually is, including a confidence score whose per-infant correlation fell from +0.710 to +0.029. Recomputing every figure from stored per-clip predictions instead of from training logs is what made them findable.

6.5 Planning and scope

1. **The schedule allowed one design iteration, not two.** The plan carried a single architecture, built once. Building it, measuring that it solved the wrong problem, diagnosing why and replacing it consumed the slack that would otherwise have enlarged the cohort (§7.4).
2. **Four items were budgeted at zero because nobody expected them.** Storage, adapters and cabling, printed mechanical parts, and peripherals. All four are consequences of moving from a design on paper to a rig in a working ward, and §7.5 recommends an explicit integration contingency instead.

3. **Two recording sources were collected and never used.** The whole of 2026-04-29 and the second patient folder of 2026-06-17, both left out at the deduplication stage and neither excluded on the basis of any result. §7.7 lists their recovery as the cheapest available increase in cohort size.

7 Conclusions and Recommendations

7.1 Measured performance against the initial specification

The declared specification had twelve elements, and they are assessed element by element below.

- **Met:** Contactless operation from NIR video alone. An executable pipeline from video in to vital signs out, end to end on the Orin. A trained and validated model, 27 of them, under leave-one-subject-out over nine infants. Ground truth from the clinical monitor by a full computer-vision pipeline, with 98.487 % of the cohort's 300,000 frames usable after quality control and the three defect counters recomputed over all 500 clips (§5.2). A validation report carrying MAE, RMSE, bias, limits of agreement and both overall and within-subject correlation, each with an infant-clustered interval. A clinical dataset of 500 clips from nine infants. And beating a classical baseline by +2.58 bpm over a calibrated peak detector, CI [+0.4, +5.2], on eight of nine infants.
- **Met in part:** The respiratory-rate target was declared as MSE at most 2.0, an RMSE of 1.41 bpm. The delivered system measures clip MAE 9.73 bpm, CI [6.9, 12.9], and clip RMSE 12.5, against a no-skill baseline of 13.6 to 14.0. Temperature estimation at MSE at most 0.2 °C was not attempted (§5.7). And the dense per-frame output was not delivered: the system emits one estimate per 10 s window, eleven per clip, which §4.7.2 defends as a consequence of what the label can support and not as a convenience.
- **Not assessable: Robustness** to blanket coverage. The condition is recorded on every clip, but all 500 clips in the analyzed cohort are marked exposed, so no covered recording ever reached the data. This is a gap in the dataset and not an untested feature, and no analysis closes it.

7.2 Strengths and weaknesses of the product

7.2.1 Strengths

- **It measures something inside each baby, not only the difference between babies.** The representation contains no spatial information and the read-out cannot use absolute position, so the appearance shortcut that killed the first architecture has no representation in the output. The evidence that something within-infant survives is a within-subject correlation of 0.288 with an interval excluding zero, roughly 8 % of the within-infant variance; a model that had merely learned each baby's average rate would show zero. §5.5.2 reports the two experiments that test this, and one went against us: identity is still decodable from the input, so the claim rests on the constraint and not on the representation being anonymous.
- **It beats a real baseline, significantly.** We compared against a calibrated classical peak detector and not a random predictor, because the detector is what a reviewer would reach for first. The delivered system beats it by 2.58 bpm with an interval excluding zero, on eight of nine infants, though §5.4 shows the interval is carried by two of them.
- **Every accuracy figure is recomputed from stored predictions.** No accuracy figure is copied from a training log. Two quantities are exceptions and are labelled where they appear: the RMSE and MSE of §5.5, which are derived from the limits of agreement and not recomputed. That discipline is what allowed us to find our own contamination and retract it.
- **The failure analysis is quantitative.** 96.3 % identity accuracy, 0.3 to 2.1 bpm collapsed spread, less spectral energy than raw pixels on 25 of 27 sampled clips. Each of these is a measurement, and together they explain the redesign instead of merely justifying it. The same discipline was then turned on the redesign, and §5.5.2 reports where it found us wrong.

- **The statistical limits are stated, not implied.** The report says what it cannot resolve: 2.17 bpm at nine infants, and 43 infants to resolve 1 bpm. That number is what makes the recommendations in §7.7 specific instead of aspirational.
- **The clinical apparatus works.** IRB approval, a multi-camera rig in a working intensive-care unit, lossless synchronized capture, a mechanical solution the staff tolerate, and a ground-truth pipeline whose three failure modes are named and masked.

The procedure grades the principal characteristics achieved, so we name them and where the evidence sits and not leave a marker to infer it.

Table 21: *Principal project characteristics, and where each is evidenced.*

Characteristic	Evidenced in
System implementation and construction	§4.2, the ward rig, mounts and capture pipeline; §6.1
Algorithmic design	§4.5 and §4.6, the spectral front end and the shift-equivariant read-out
Mathematical analysis	§4.7.2, the translation argument; §5.1 and §5.6, the estimators, the clustered bootstrap and the power analysis
Integration with an existing system	§4.3, the bedside monitor as reference; §4.4, third-party MIT magnification; §6.1 item 5, the ward acquisition service
Performance research	§5.4 to §5.6, leave-one-subject-out, confidence intervals, ablations and refuted claims

7.2.2 Weaknesses

- **It is not clinically accurate.** Limits of agreement of -25.4 to +23.6 bpm, and only 37 % of clips within 5 bpm. It is a proof of concept that the signal is recoverable, not a device. §5.4 measures the coverage-for-accuracy trade that is the nearest route to a usable bedside mode, with its three caveats, and §7.7 recommends building and validating it prospectively.
- **Predictions are compressed.** A slope of 0.40 means the model moves less than half as far as the true rate does. Of the three candidate causes in §5.5, label attenuation is ruled out arithmetically and read-out shrinkage by measurement, so what remains is the evidence reaching the network. That is the harder of the three to fix and it is where §7.7 points.
- **Nine infants is too few to design with.** Nothing below 2.17 bpm is resolvable, and every architectural variant tried landed inside that band. The design questions that remain cannot be settled at this cohort size.
- **The dense-output requirement is unmet.** The specification asked for 200 values per clip and the system produces 11. The defense in §4.7.2 is sound but it is a defense.
- **There are no covered recordings in the cohort.** All 500 clips are marked exposed, so the data holds no example of the commonest obstruction in a NICU. That is a gap in the dataset and not in the model, and no amount of analysis closes it.
- **The input is not anonymous.** A linear classifier still names the infant from the (6, 160) spectra on 85.5 % of held-out clips, against an 18.5 % floor and 93.2 % once the rate is held fixed (§5.5.2). The read-out cannot use that, and the measured symptom of the first architecture is gone, but the design argument is weaker than the one §4.6 originally made and we state it as such.
- **Temperature was not delivered.** Half of the original two-branch goal is absent.
- **The published repository is misleading.** It still presents the abandoned architecture as the system, and the delivered pipeline lives outside it.

7.3 The project goals against what was achieved

Partly, and asymmetrically. The project had two halves: build a clinical acquisition and ground-truth system, and build a model that turns its output into vital signs. The first was achieved in full and is

arguably the harder half. The second produced a working system whose accuracy falls well short of the declared target.

The declared RR target was an MSE of at most 2.0, which read strictly means an RMSE of 1.41 bpm. We reached a clip MAE of 9.73 bpm and a clip RMSE of 12.5 bpm. It is worth recording how ambitious that target was: the best published mean absolute error on real infant video is about 2.9 bpm on the AIR-125 benchmark [12], against 4.7 to 6.2 bpm for models designed for adults, and the 1.6 bpm figure sometimes quoted comes from thermal video [13], a modality that images the respiratory signal far more directly than NIR does. The target set in the preliminary report was therefore stricter than the published state of the art on the modality we actually use. That does not make 9.73 bpm good, but it does place it.

A second point matters more. The target was written as an MSE of at most 2.0 bpm^2, an RMSE of 1.41 bpm, and the reference we scored against is a photograph of a monitor whose displayed rate varies by 3.2 bpm within a 10 s window (§4.3.2). Treating that as reference error, no model could measure below about 3.2 bpm RMSE against this reference however good it became, so a target of 1.4 could not have been distinguished from a target of zero: as specified it was unfalsifiable and not merely hard. Two qualifications belong with that. The within-window figure is an upper bound on the reference error. And the floor explains almost none of our shortfall: subtracting 3.2 bpm in quadrature from our 12.5 bpm RMSE leaves 12.1 bpm of model error, so the reference accounts for about 7 % of our mean squared error. The lesson is about order of work: a target should be set against a reference already characterized, and §7.7 recommends that to anyone repeating this.

What the project did achieve is the following. A camera-only pipeline on real NICU recordings of nine preterm infants carries information about respiratory rate that beats both a no-skill predictor and a calibrated classical detector under leave-one-subject-out validation, and it does so through a representation in which patient identity is not available as a shortcut. Establishing that required first building a system that failed, and then diagnosing the failure precisely enough to know what to change.

That sequence is the part we consider the real result, and the case for it is not ours alone. Lipton and Steinhardt put it in one sentence: empirical study aimed at understanding can be illuminating even absent a new algorithm [46]. They also name the failure to identify the sources of empirical gains as a troubling trend, citing a re-evaluation in which a lineage of published architectural improvements turned out to be hyperparameter tuning. That is the trap §5.6 says this cohort is too small to avoid by A/B testing, and why §7.7 recommends adding infants instead of variants. The redesign was not a guess: it followed the torso-motion diagnostic of §4.8, which pointed at motion measures and spectral peak finding.

7.4 Plan versus execution: schedule

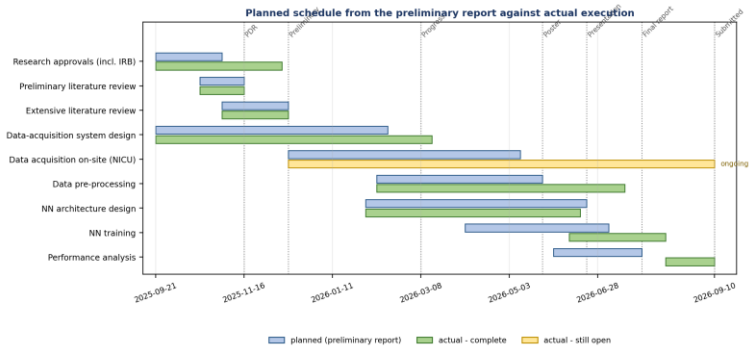


Figure 11: *Planned schedule from the preliminary report against actual execution. The planned bars are taken from the preliminary report. The actual bars are reconstructed from the progress report and the repository commit history, so they are accurate to the week instead of the day, and tasks still open at submission are marked ongoing. [[PH-20]]*

Table 22: *Planned duration against outcome, per task.*

Task	Planned (weeks)	Outcome	Comment
Research approvals	6	Overran	IRB approval set the earliest possible recording date and was not under our control
Preliminary literature review	4	On schedule	
Extensive literature review	6	On schedule	Delivered in the preliminary report
Data-acquisition system design	21	Overran	Absorbed the three hardware failures of §6.1
Data acquisition on-site	21	Continuing	Paced by the ward. Nine infants recorded so far
Data pre-processing	15	Complete for the delivered pipeline	Monitor OCR, label QC, magnification and spectral feature caching all in place
NN architecture design	20	Complete, twice	The first design was built, measured and replaced. The second is the delivered system
NN training	13	Complete	27 fold-seed models under LOSO
Performance analysis	8	Complete	This is §5, including the audit that voided the earlier results

Four lessons concern the plan itself and not its execution:

- **The dependency was under-modeled.** The Gantt placed training in parallel with pre-processing, but training cannot start before magnified clips and quality-controlled labels exist. In practice the two were in series.
- **Clinical data rate is not a project variable.** The plan assumed a steady arrival of recordings once the system was deployed. A clinical acquisition schedule should be planned as a rate with an interval, not as a duration.
- **The March forecast was wrong.** The progress report judged the early hardware delay minor and the project aligned for its final milestones. Two causes ran against that. The first was the security situation of March 2026, which restricted access to the hospital and to the campus and held up both recording and development at the point the forecast was being made. The second is that the architecture had to be built twice, and this report is submitted six and a half weeks after the planned date of 2026-07-26. The slip is attributable to the rebuild described in §5.3 and §6.3, and we did not see it coming in March because the first architecture was still training and had not yet been diagnosed.
- **The schedule allowed one design iteration, not two.** The plan carried a single architecture, built once. Building it, measuring that it solved the wrong problem, diagnosing why, and replacing it consumed the slack that would otherwise have gone into enlarging the cohort. A schedule for a project whose first architecture is unproven should carry an explicit iteration allowance, in the same way §7.5 recommends an explicit integration contingency on cost.

7.5 Project cost

Table 23: *Planned cost by item.*

Item	Qty	Planned unit (USD)	Planned total (USD)
IR camera	1	285	285
e-con CU30 (NIR, patient)	1	152	152
e-con e-CAM82 (vital-signs monitor)	1	110	110
e-con CU27 (temperature monitor)	1	200	200
NVIDIA Jetson AGX Orin dev kit	1	2,500	2,500
Camera stands	2	10	20
Planned subtotal			3,267
1 TB NVMe SSD (not budgeted, §6.1 item 2)	1		0
USB3 adapters and cabling (not budgeted)			0
3D-printed arms and connectors (not budgeted, §6.1 item 4)			0
Mouse, keyboard, screen			0

Total			3,267

The instructive part of this table is the block of unbudgeted rows. All four are consequences of moving from a design on paper to a rig in a working ward: storage sized for real video volumes and not test clips, adapters for the actual connector set, and mechanical parts we had to design because no commercial mount met the clinical constraints. A future proposal for a comparable project should carry an explicit integration contingency instead of budgeting only the named components.

7.6 Milestones

Table 24: *Project milestones. Every milestone except the final report was delivered on its planned date.*

Milestone	Planned date	Actual date
PDR	2025-11-16	2025-11-16
Preliminary report	2025-12-14	2025-12-14
Progress report and presentation	2026-03-08	2026-03-08
Poster submission	2026-05-24	2026-05-24
Conference presentation	2026-06-21	2026-06-21
Final report	2026-07-26	2026-09-10

Every scheduled deliverable except the final report was submitted on its planned date. The final report slipped by six and a half weeks, and the rest of the schedule slipped inside the project, which §7.4 reports task by task.

7.7 Recommendations for continuation

Ordered by expected value against effort. The first follows directly from §5.6 and the second from the slope in §5.5. Four things a continuation might otherwise start with are already done and are reported as measurements: the cohort-wide label-QC recompute (§5.2), the within-infant prediction spread (§5.5.1), the video-to-monitor latency (§4.3.2), and the timing of the pipeline (§3.5).

1. **Enlarge the cohort before anything else.** At nine infants nothing below 2.17 bpm is resolvable and 43 infants would be needed to resolve 1 bpm, so every design question that remains open is open because the cohort cannot answer it. This is the one recommendation that unblocks the others, and its corollary is that starting a new architecture on the present cohort would produce a different number and no knowledge, because the interval is too wide for the change to be detected either way.
2. **Attack the compression.** Of the three causes in §5.5, label attenuation is ruled out arithmetically and read-out shrinkage by measurement: all three slope intervals in Table 15 exclude unity. That leaves the evidence reaching the network, so the levers are region-of-interest placement, the magnification amplification factor and pass-band, and the window length, all cheap to sweep because the spectra are cached. The median read-out scores 8.48 bpm against 9.26 for a one-line change, but its interval crosses zero, so re-measure it on the enlarged cohort instead of treating it as a gain.
3. **Evaluate on a public benchmark.** The AIR-125 [12] and AIR-400 [47] infant-respiration benchmarks would place the 9.73 bpm reported here on the same axis as published work, at no clinical cost. Published mean absolute error on AIR-125 is about 2.9 bpm, which is the gap to close.
4. **Probe any future read-out before believing its accuracy.** §5.5.2 measures that the infant is still identifiable from the model's own input at 85.5 %, so the present system is protected by the constraint on its read-out and not by an anonymous representation. Any successor with more freedom, and a transformer over these spectra is the obvious candidate, can re-acquire the shortcut. Run the probe of §5.5.2 and the within-infant spread of §5.5.1 alongside the accuracy, and treat an accuracy gain that arrives with a collapse in within-infant spread as the shortcut returning.
5. **Drop the frozen ResNet-50 from the deployed pipeline.** §5.5.2 measures the retrain on channels 1 to 3 alone at 9.25 bpm against 9.26, a paired difference of -0.01 with the whole interval inside the

resolvable band. Removing it deletes the feature-caching stage and the 875 MB `feats_9inf.npz` from the deployment path for no measured accuracy cost. Carry the caveat with it: three infants get worse, one by 6.5 bpm, and nine infants cannot resolve that either way.

6. **Recover the two recording sources already on disk.** The whole of the 2026-04-29 session, 63 clips from one patient, and the second patient folder of 2026-06-17 exist and contribute nothing to the cohort (§4.2). Bringing them in raises the cohort by roughly a fifth for the cost of running the existing extraction pipeline, with no new clinical work. Enter the second 2026-06-17 patient under its own `subject_id` and check neither source already appears in the cohort under another name, because that is exactly how `legacy_P0` became a duplicate (§6.4).
7. **Record covered infants.** Every clip in the present cohort is marked exposed, so the data holds no example of the commonest occlusion in a NICU. There is no split to run on the existing results, only new recording to do. This is the second data-collection priority after enlarging the cohort.
8. **Build the confident-only reporting mode.** §5.4 shows the classical detector reaching 3.66 bpm on the sharpest 30 % of clips using a gate that needs no label, against a 14.89 bpm no-skill baseline on that subset. Pair it with the always-on CNN estimate instead of choosing between them, and validate the gate prospectively and not on the same clips that set the threshold.
9. **Finish characterizing the reference.** The video-to-monitor latency is now measured at half a second and constant (§4.3.2), so what remains is the value-level audit: read a sample of frames by hand against the stored labels and report the residual error in breaths per minute. Re-extract `0617_P0` first, since its latency and its masked-frame count are both anomalous.
10. **Report accuracy stratified by skin tone and gestational age.** Both are already recorded per session. Optical vital-sign measurement is known to degrade on darker skin, and that should be measured and not hoped away.
11. **Measure what the magnification is worth, then replace it.** Cache the same cohort from the unmagnified crops, which `tools/cache_windowed.py` supports through `--metadata`, and run it on channels 1 to 3, since §5.5.2 shows those alone match the full six and the pixel-block spectra are the only place magnification can act. One variable, about 50 minutes. §4.4 measures the stage amplifying out-of-band motion more than the respiratory band on 16 of 61 clips, and the two demonstration videos show it changes the picture, not the answer. If it earns its place, replace it instead of tuning it: the network dates from 2018 and only the caching step has to re-run to evaluate a successor.
12. **Revisit dense output only when the label supports it.** A per-frame output becomes meaningful if a reference with per-frame resolution is obtained, for example a synchronized digital feed from the monitor instead of a photograph of its screen.

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Appendix A: Repository map and how to run the pipeline

The delivered pipeline lives on the Orin, outside the public repository. Both locations are given, because a reader of the repository alone would reconstruct the abandoned architecture (§7.2.2).

```
# ---- delivered pipeline (Orin) ----------------------------------------
/mnt/pag_disk/rr_training/
  tools/train_logfreq.py         ShiftEquivariant model + spectral dataset
  tools/train_windowed.py        LOSO driver, --fixed-epochs N (0 = nested ES)
  tools/estimate_rr.py           classical peak detector + calibration, --cal-out
  tools/confidence.py            per-clip confidence score (see 5.6: weak)
  tools/scan_9inf.py             cohort scan over the clean 9-infant set
  tools/blend_loso.py            detector/CNN blending experiment (see 5.6: no gain)
  tools/cache_windowed.py        stage 4: blocks, FFT, whitening, log resample
  tools/review_stats.py          all reported statistics, from stored predictions
  tools/probe_identity_spectra.py  identity probe on the (6,160) spectra (5.5.2)
  tools/probe_identity_control.py  the rate-matched control for it (5.5.2)
  tools/readout_compare.py       expectation vs argmax vs median (5.5, Table 15)
  tools/slice_spec_channels.py   builds the channels-1-to-3 dataset (5.5.2)
  tools/make_plot_data.py        builds plot_data.json for the figures
  tools/stamp_checkpoint.py      writes measured metrics into a checkpoint
  data/metadata_9inf_dedup.csv   THE cohort file: 500 clips, 9 infants, deduplicated
  runs/31_windowed/              reference configuration
  runs/40_fixed150_loso/         fixed-150-epoch comparison
  runs/41_curves150/curves.csv   per-epoch curves, 4050 rows
  runs/50_ch123/                 channels 1-3 only, fixed 150 (Table 19)
  runs/51_dist/                  stored output distributions + read-out comparison
  demo/                          the two demonstration videos of 5.5.1
  analysis_2026_08_27_review/    ba_points.csv, per_infant_mae.csv, review_stats.json,
                                 plot_data.json, scan_9inf.csv, blend_*.csv
  backups/<name>_<ts>/           snapshots with RESTORE.sh; backups/LATEST

/mnt/pag_disk/deepmag_work/      DeepMag magnification outputs
/mnt/pag_disk/Test_Records_ROIs/ ward recordings: <date>/Patient_N/Record_M/vid_*/
                                 motion_frames, chromatic_frames, signs/rr.npy

# ---- public repository (mostly the abandoned design) ------------------
Final_Project_Algo/
  macro_motion_rr_package/
    scripts/build_metadata_from_magnified.py    STILL LIVE: label QC
    macro_motion/utils/seed.py                  STILL LIVE
    macro_motion/models/*.py                    abandoned ResNet + transformer
    configs/*.yaml                              abandoned configuration
  deep_motion_mag/                              third-party MIT magnification (TF 1.x)
```

A.1 Commands

```
# environment
cd /mnt/pag_disk/rr_training
export LD_LIBRARY_PATH=/mnt/pag_disk/rr_venv/syslibs
export PYTHONPATH=/mnt/pag_disk/rr_training/tools

# magnification (TensorFlow 1.x environment, from deep_motion_mag/)
sh run_temporal_on_test_videos.sh \
   o3f_hmhm2_bg_qnoise_mix4_nl_n_t_ds3 <video_name> 8 0.2 2.0 30 1 butter

# leave-one-subject-out training, nested early stopping
/mnt/pag_disk/rr_venv/bin/python tools/train_windowed.py \
    --metadata data/metadata_9inf_dedup.csv --out runs/31_windowed

# same, with a fixed epoch budget instead of early stopping
/mnt/pag_disk/rr_venv/bin/python tools/train_windowed.py \
    --metadata data/metadata_9inf_dedup.csv --fixed-epochs 150 \
    --out runs/40_fixed150_loso

# every statistic in chapter 5, recomputed from stored predictions
/mnt/pag_disk/rr_venv/bin/python tools/review_stats.py
```

One operational rule governs all of the above: the ward acquisition service outranks this pipeline. It must never be killed or renamed to free resources; the training job is deprioritized instead.

Appendix B: Grade Recommendation Page (blank)

Reproduced from section 12.3 of the 2026 Engineering Projects Procedure, to be completed by the supervisor. Scoring scale: 1 weak (20), 2 average (40), 3 good (60), 4 very good (80), 5 excellent (100).

<u>המלצת ציון לדו״ח מסכם</u>

אם יש צורך, לכל סטודנט/ית בנפרד

מספר הפרויקט: p-2026-106 שם הפרויקט: MagViVit

שם המנחה: _____

שם הסטודנט/ית: _____ ת.ז.: _____

%	קריטריון	1 - חלש	2 - בינוני	3 - טוב	4 - ט״מ	5 - מצוין
9	הגדרת המטרה - האם מטרת הפרויקט ברורה? המטרה צריכה לכלול גם את התרומה הצפויה מהשלמת הפרויקט.					
10	ארכיטקטורת הפתרון - האם הובהר מדוע הארכיטקטורה המוצעת לפתרון מתאימה לפתרון הבעיה? האם הייתה התייחסות לאלטרנטיבות?					
9	הצגת התוצאות - האם הוסברו המשמעויות של התוצאות, או המסקנות הנובעות מהן, האם רק הייתה הצגה עובדתית של תוצאות?					
9	מבנה - מצגת - האם יש מבנה הגיוני להצגת העבודה? מבוא, רקע ו-SoTA, מטרות, שיטה, פתרון, תוצאות, מסקנות.					
9	רמת קושי - כיצד הנך מעריך את הפרויקט בהשוואה לפרויקטים אחרים השנה או בשנים קודמות?					
9	איכות כתיבה / הצגה - נא להתייחס לאיכות הצגת העבודה: תחביר, מבנה וקשר הגיוני בין החלקים במהלך הצגת העבודה.					
9	שורה תחתונה - האם מטרות הפרויקט הושגו? יש להתייחס למטרות כפי שהוגדרו בתחילה.					
9	הבנה - האם נראה שהסטודנט מבין את העבודה שנעשתה? האם עומק הדיון מספק?					
9	השקעה - האם ניכר שהסטודנט השקיע בעבודה בפרויקט ו/או בהכנת ההצגה של העבודה? האם נראה שהעבודה נעשתה בחופזה, כדי לצאת ידי חובה?					
9	דירוג - כיצד הנך מעריך את איכות הפרויקט ביחס לכלל הפרויקטים שראית?					
9	המאפיינים העיקריים בפרויקט - יש לסמן את כמות המאפיינים שהושגו (ולו באופן חלקי). מאפיין אחד בלבד יזכה במחצית הניקוד. שני מאפיינים ומעלה יזכו בניקוד מלא. המאפיינים: * מימוש/יצור מערכת, * אנליזה מתמטית, * תכנון אלגוריתמי, * אינטגרציה עם מערכת קיימת, * חקר ביצועים.					

הערכת רמת הקושי של הפרויקט: קל מאוד / קל / בינוני / קשה / קשה מאוד

אם יש כוונה לפרסם/ פורסם מאמר, שם כתב העת ומועד משוער להגשה: _____

ציון אם יש כוונה לשקול המלצה כפרויקט מצטיין: _____

הערות נוספות: _____

תאריך : _____ חתימה : _____